



AI integration for climate risk mitigation: The role of organizational context

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ABSTRACT

This study examines the relationship between artificial intelligence (AI) adoption and Firm-Level Climate Change Risk (FLCCR) among Chinese enterprises. Using comprehensive firm-level data on AI implementation and FLCCR exposure, we analyze the contextual effectiveness of AI across diverse ownership structures, industry sectors, and corporate governance frameworks. Our empirical analysis reveals a robust association between AI adoption and reduced FLCCR, with findings consistent with established economic theories. The results remain statistically significant after addressing potential endogeneity concerns through multiple robustness checks. Our findings reveal that AI's climate risk-reduction potential is not uniform but context-dependent, varying significantly across ownership types, sectors, and governance characteristics. Notably, the risk-mitigating effects of AI appear particularly pronounced in state-owned enterprises, firms operating in pollution-intensive or high-technology sectors, and organizations with strong corporate governance mechanisms, specifically those characterized by board independence and gender diversity. These findings contribute to the growing literature on technological solutions for environmental challenges while providing actionable insights for corporate decision-makers and policymakers seeking to enhance climate resilience through strategic AI integration. The study underscores the potential role of AI as a tool for sustainable development while acknowledging the complex interplay between technological adoption and organizational factors in risk mitigation outcomes.

1. Introduction

Climate change, defined as “exposure and vulnerability to the physical, transition, and regulatory risks associated with climate change,” is one of the most pressing challenges of the 21st century, with its cascading effects threatening global economic stability and corporate resilience (Sautner et al., 2023). Businesses today face unprecedented risks—from extreme weather events to regulatory pressures—demanding innovative solutions to navigate this complex landscape (Li et al., 2024). Corporations navigate a complex landscape of climate-related risks, encompassing physical, transitional, and regulatory dimensions. These risks are generally classified under the umbrella term of Firm-Level Climate Change Risk (FLCCR). Physical risks manifest as direct climate impacts, such as extreme weather events and rising sea levels, jeopardizing assets and operations. Transitional risks stem from the

imperative to decarbonize business models, necessitating significant investments and operational shifts. Moreover, regulatory risks arise from evolving climate policies, imposing potential compliance burdens and business costs. This intricate interplay of climate-related challenges underscores the importance of understanding and managing these risks for corporate resilience.

The economic implications of climate change have emerged as a critical area of inquiry in recent decades. While initially met with skepticism, the scientific consensus on anthropogenic climate change has solidified, prompting increased attention to its economic consequences. Pioneering work by Nordhaus (1977) laid the groundwork for subsequent research, with scholars such as Kelly and Kolstad (1999) and Nordhaus and Popp (1997) contributing to the evolving discourse on the economic costs of climate change. The Kyoto Protocol in 1997, the Stern Review in 2006, and the Paris Agreement in 2015 have further

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galvanized global efforts to address climate change, culminating in heightened scrutiny of its impact on corporate performance (Bolton and Kacperczyk, 2021; Chava, 2014; Dessaint and Matray, 2017; Hong et al., 2019; Huynh and Xia, 2021; Javadi and Masum, 2021; Painter, 2020). Moreover, the expanding regulatory landscape and heightened investor expectations have amplified the pressure on firms to manage Climate Change Risks (CCR) effectively.

Amidst climate-related challenges, artificial intelligence (AI) has emerged as a transformative force capable of reshaping how firms understand, mitigate, and adapt to climate risks. By leveraging AI, organizations can more effectively navigate the complex and interrelated domains of physical, transitional, and regulatory climate risks, thereby enhancing their resilience and sustainability. While AI adoption has been empirically linked to improvements in economic growth (Aghion, 2018), productivity (Czarnitzki et al., 2023), financial performance (Bosse et al., 2023), and innovation (Bahoo et al., 2023), its potential role in advancing environmental sustainability is gaining increased scholarly attention (Hussain et al., 2022). However, despite these promising developments, the specific mechanisms through which AI mitigates firm-level climate change risk—also called climate change liability (CCL)—remain underexplored. This represents a significant gap in both theoretical and practical understanding. As the intersection of digital technology, environmental responsibility, and corporate governance becomes increasingly critical (Hickman and Petrin, 2021; Liu et al., 2022), AI is being positioned as a key enabler of sustainable development across sectors (Goralski & Tan, 2020; Leal Filho et al., 2024). This study aims to bridge the gap by empirically investigating the role of AI adoption in mitigating firm-level climate change risk, offering novel insights into how technological innovation can support sustainable and accountable business practices.

Although prior research has explored the role of AI adoption in addressing climate change risks, most existing studies have taken a conceptual, review-based, or case-study approach. For example, Boero (2024) conducted an AI-enhanced systematic review of climate adaptation costs, while Ai and Gao (2023) examined the impact of extreme climate events on firm-level risks. Similarly, Jones et al. (2023) illustrated AI's potential in flood-related climate applications, and Akomea-Frimpong et al. (2023) synthesized 36 studies on AI's role in climate risk management within public-private partnerships. Jain et al. (2023) commented on emerging AI innovations for climate adaptation. In a parallel stream, several studies have assessed the impact of firm-level climate change risk on various financial, accounting, and operational outcomes, such as accounting conservatism (Ferdous et al., 2024), tax avoidance (Song and Xian, 2025), corporate cash reserves (Tran et al., 2025), debt maturity (Goodell et al., 2025), cost of capital and investment efficiency (Agoraki et al., 2024), total factor productivity (Ren et al., 2022), CEO equity incentives (Hossain et al., 2025), and firm efficiency (Hoang et al., 2025). However, to our knowledge, no study has yet investigated a technology-based mechanism, such as AI adoption, for mitigating firm-level climate change risk. Thus, our study establishes a new domain of literature to discuss the technology-based solutions to reduce firm-level climate change risks.

Our study contributes uniquely to this literature in three key ways: (1) Unlike prior work that primarily employs reviews, commentaries, or qualitative case studies, we offer large-sample empirical evidence based on panel data integrating financial and environmental reporting; (2) We are among the first to examine the direct effect of AI adoption on firm-level climate change risk; and (3) We introduce novel moderating factors—corporate governance characteristics, ownership structure, and industry classification—to assess how these organizational dimensions shape the effectiveness of AI in mitigating climate risk. This study advances theoretical understanding by elucidating the nuanced relationship between AI deployment and climate-related risks. It establishes a robust framework integrating environmental economics, organizational theory, and information systems to comprehensively examine the intricate interplay between technology, governance, and sustainability

outcomes. Furthermore, the study extends existing theoretical boundaries by investigating the differential impacts of AI implementation across diverse organizational contexts, such as ownership structures and industry sectors. This research significantly contributes to our comprehension of the complexities inherent in integrating technological advancements into sustainability strategies, emphasizing the critical role of contextual factors in shaping the effects of AI adoption on climate change risk management.

This study delves into the intricate relationship between AI adoption and firm-level climate change risk, examining the potential of AI to serve as a strategic tool for environmental sustainability. We hypothesize a negative association between AI adoption and firm-level climate change risk, suggesting that increased AI implementation can reduce such risks. To further understand this relationship, we investigate the moderating effects of firm ownership (state-owned vs. private), industry characteristics (high-tech vs. polluting), and corporate governance (board independence, gender diversity). By comparing AI adoption patterns and climate change risk outcomes across these dimensions, we aim to identify effective strategies for leveraging AI to address environmental challenges.

China aspires to lead the world in AI technology and gain geopolitical and economic advantages from this crucial technological domain (Khanal et al., 2025). China's sectors are turning to AI technology more widely, which has improved processes and given companies a competitive edge (Roberts et al., 2021). In several industries, such as manufacturing, financial services, health care, and e-commerce, artificial intelligence (AI) systems such as computer vision, natural language processing, and algorithmic learning have been deployed (Barton et al., 2017). For Beijing, artificial intelligence (AI) is a vital engine of economic growth and a way to outperform other countries. Intending to encourage the broader application of AI, the Chinese government has started to grant support, tax credits, and financing for the development of artificial intelligence (AI).

The Communist Party of China's 20th National Congress Report emphasizes the significance of ecologically sound development in promoting economic progress (Jinping, 2022). The report's findings encourage using a comprehensive protection strategy, developing carbon-neutral and environmentally responsible industries, and providing incentives for using low-carbon and sustainable manufacturing methods. China has established aggressive targets to reduce emissions by 2030 and become carbon neutral by 2060 (Dinga and Wen, 2022; Zhang and Hanaoka, 2021), forcing a switch to green energy sources such as wind and solar power (Jia and Lin, 2021). According to the administration, mitigating and adapting to climate change is essential to maintaining social wellness and guaranteeing the integrity of the current political order. China's simultaneous investment efforts in artificial intelligence and climate change risk reflect its ambitions for global leadership in technology (Lundvall and Rikap, 2022) and responsible environmental behavior (Abbasi and Erdebili, 2023). AI offers a route to technical domination and financial success, but tackling risks associated with climate change is a question of survival and preserving the country's social harmony and ongoing economic growth. China's efforts in artificial intelligence (AI) and climate change resistance will significantly impact the global landscape in the following decades as the country strives to negotiate the delicate balance between environmental sustainability and technological growth (Cheng and Zeng, 2023; Feng et al., 2024; Khanal et al., 2025).

The empirical evidence presented in this study reveals a clear inverse relationship between adopting artificial intelligence (AI) at the firm level and associated climate change risk. This suggests that AI technologies can be a valuable tool for mitigating environmental impacts. Furthermore, the study demonstrates that the effectiveness of AI in reducing climate change risk is more pronounced in state-owned enterprises (SOEs) and firms operating within polluting industries. Additionally, the study underscores the importance of corporate governance, with higher levels of board independence and gender diversity

associated with a stronger negative relationship between AI adoption and climate change risk.

The rest of the draft presents the literature review, the methodology adopted, empirical results, discussion, and ultimately conclusion, along with limitations, recommendations, and implications of results.

2. Literature review and hypotheses development

To develop the hypotheses, a structured literature search was conducted using Scopus, Web of Science, and Google Scholar databases. Keywords such as “AI adoption,” “digitalization,” and “firm-level climate change risk” were used to formulate the baseline hypotheses. Peer-reviewed articles published within the last ten years were initially prioritized; however, the time frame was extended where necessary to gain a deeper understanding of the mechanisms through which AI adoption may influence climate change risk. Studies were selected based on their empirical relevance and theoretical contribution to the relationship between AI and climate-related firm outcomes, particularly in the context of risk mitigation and sustainability practices.

Following the development of the baseline hypotheses, the study further examined the moderating effects of firm characteristics, including ownership structure (state-owned vs. non-state-owned), industry classification (high-tech vs. non-high-tech; polluting vs. non-polluting industries), and corporate governance attributes (firms with high vs. low board independence; and firms with high vs. low board gender diversity). To support these extended hypotheses, the literature search was broadened to incorporate additional keywords related to ownership, industry typologies, and corporate governance mechanisms, ensuring a comprehensive theoretical foundation.

2.1. Artificial Intelligence and climate change risk

Real-world implementations demonstrate AI's climate mitigation potential across sectors. In energy, using neural networks, Google's DeepMind reduced data center cooling costs by 40 % (Li et al., 2019). Some countries are developing AI-based irrigation systems to control drought. Manufacturing benefits from Siemens' AI-driven waste reduction systems, decreasing material use by 17 % in automotive plants. These cases confirm AI's adaptability while highlighting measurable environmental benefits, though outcomes depend on sector-specific constraints like data availability and infrastructure readiness. While recent surveys (e.g., Akomea-Frimpong et al., 2023; Boero, 2024) explore AI's role in climate risk, they focus on sector-specific or conceptual insights, neglecting firm-level empirical analysis. Unlike these works, our study provides large-scale evidence on how AI mitigates climate risk, while uniquely examining governance, ownership, and industry moderators—gaps overlooked in prior reviews. This advances the field beyond theoretical discussions, offering actionable insights for organizational sustainability.

Artificial intelligence (AI) holds significant potential to mitigate firm-level climate change risks through its capacity to optimize resource use, enhance predictive capabilities, and improve operational efficiency. AI technologies enable firms to reduce their environmental impact by identifying low-carbon alternatives across supply chains (Singh and Goyal, 2023), optimizing energy consumption (Ahmad et al., 2022), and modeling climate scenarios (Dannouni et al., 2023). Specific applications range from real-time emissions monitoring (Tseng and Lin, 2024) to predictive analytics for extreme weather events (Galaz et al., 2021), with studies suggesting AI could contribute to a 4 % reduction in greenhouse gas emissions across industries through efficiency gains (Dannouni et al., 2023). These capabilities align with multiple Sustainable Development Goals, particularly SDG 13 (Climate Action) through intelligent resource management (Singh et al., 2024), and SDG 12 (Responsible Consumption) via waste reduction. However, the climate benefits of AI adoption are not without limitations. Significant energy demands from data centers (Dauvergne, 2022) and

environmental costs associated with hardware production (Katal et al., 2023) present notable trade-offs. Implementation barriers also exist. These constraints suggest that AI's net impact on climate risk depends on how firms deploy these technologies and offset their environmental costs.

The Resource-Based View (RBV) provides a theoretical framework to explain variations in AI's climate risk mitigation effects across firms. According to RBV, AI capabilities constitute valuable, rare, and imperfectly imitable resources that can generate sustained competitive advantages (Bendig et al., 2023). Firms that effectively integrate AI into their strategic resources gain superior capabilities for climate risk management through three mechanisms: (1) enhanced data processing for emissions tracking and reporting (Nishant et al., 2020), (2) predictive analytics for physical climate risks (Jain et al., 2023), and (3) optimization of energy-intensive processes (Rana et al., 2022). These AI-driven capabilities are particularly valuable for climate risk reduction because they address mitigation (through efficiency gains) and adaptation (through predictive modeling) needs.

This theoretical foundation leads to our first hypothesis: the negative relationship between AI adoption and firm-level climate change risk emerges because AI capabilities enable firms to (a) reduce emissions through operational efficiencies, (b) anticipate and adapt to physical climate impacts, and (c) meet stakeholder demands for climate accountability - with these effects being most substantial when AI systems are rare, valuable, and well-integrated into firm resources. Thus, we propose:

Hypothesis 1. There is a negative relationship between AI adoption and firm-level climate change risk, as AI capabilities provide firms with unique resources to mitigate emissions, enhance adaptation, and improve climate risk management.

2.2. Role of state-ownership

State-owned enterprises (SOEs) operate under unique institutional conditions that systematically amplify the relationship between AI adoption and climate change risk (CCR) mitigation. Drawing on the natural resource-based view (NRBV) (Pan et al., 2020), SOEs are incentivized to leverage AI for sustainability due to their mandate to align with national policy objectives (Spagnuolo et al., 2025). Unlike private firms, SOEs face heightened institutional pressures—including stringent regulatory oversight, public accountability, and political directives (Lu et al., 2024)—that compel them to prioritize AI-driven CCR reduction as part of their societal obligations (Zhang et al., 2022).

The centralized governance structures of SOEs play a crucial role in enabling rapid deployment of AI technologies for environmental monitoring and compliance, while their privileged access to state-backed resources (Lin et al., 2024) helps overcome common adoption barriers such as financial constraints and technological readiness that typically challenge private firms (Rana et al., 2022). This institutional configuration creates a distinct moderating mechanism through three interconnected pathways. First, the policy-driven nature of SOEs leads to deeper integration of AI into climate change risk management as they strive to comply with top-down sustainability mandates (Wang et al., 2024b). Second, state support's substantial resource advantage enables SOEs to implement comprehensive AI solutions, including smart grids and predictive emissions modeling, which generate measurable reductions in climate change risks (Ansari et al., 2022). Third, the legitimacy imperative stemming from intense public scrutiny compels SOEs to demonstrate AI's effectiveness in climate risk mitigation actively, reinforcing this relationship (Petrella et al., 2021). In stark contrast, private firms' AI adoption patterns are frequently constrained by profit-maximization motives and decentralized governance structures (Kurniawan et al., 2023), factors that substantially weaken the connection between AI implementation and climate change risk reduction. Thus, we hypothesize:

Hypothesis 2. The negative relationship between AI adoption and firm-level climate change risk is more pronounced in state-owned enterprises (SOEs) than in non-SOEs, due to SOEs' institutional mandates, resource advantages, and centralized decision-making.

2.3. Role of polluting industries

Stakeholder theory posits that firms face differential stakeholder pressures based on environmental impact (Jamali, 2008). By their greater negative externalities, polluting firms are subject to heightened scrutiny from regulatory bodies, environmental activists, and communities (Albitar et al., 2023; de Grosbois and Fennell, 2022). This creates a distinct institutional context where AI adoption is a technological upgrade and a legitimizing mechanism to mitigate stakeholder concerns. The moderating effect arises precisely because polluting firms face stronger coercive and normative pressures to demonstrate environmental accountability than their environmentally friendly counterparts (Hussain et al., 2022).

The unique stakeholder pressures on polluting firms create specific mechanisms through which AI adoption more substantially reduces climate risk. First, regulatory stakeholders impose stringent emissions monitoring requirements (Delmas and Toffel, 2004), making AI-driven real-time data analytics critical for compliance. Second, societal stakeholders demand greater transparency, prompting polluting firms to leverage AI for sustainability reporting and predictive risk modeling (Galaz et al., 2021). Third, investor stakeholders increasingly penalize carbon-intensive firms, incentivizing AI adoption for emissions reduction to maintain access to capital (Bu and Shi, 2021). These mechanisms are less salient in environmentally friendly firms, where stakeholder expectations align with sustainability goals.

Furthermore, the reputational risks that pollute firms face amplify AI adoption's marginal benefit. As Dauvergne (2022) notes, high-emission sectors face disproportionate brand risks from climate activism, making AI-enabled green innovation a strategic necessity rather than an optional efficiency tool. Process optimization through AI (Xiang et al., 2021) directly addresses the core operational vulnerability of polluters—their emissions intensity—whereas in low-pollution firms, AI may only offer incremental improvements. Thus, the moderating effect stems from the greater materiality of AI's climate risk mitigation capabilities in polluting industries.

Hypothesis 3. The negative relationship between AI adoption and firm-level climate change risk is stronger in polluting firms than in environmentally friendly firms, due to heightened stakeholder pressures, regulatory scrutiny, and reputational risks that amplify the marginal impact of AI on emissions reduction and compliance.

2.4. Role of high-tech industries

The Technology-Organization-Environment (TOE) framework provides a robust theoretical lens to substantiate why the AI-climate change risk (CCR) relationship should be more pronounced in high-tech industries compared to other sectors. At the technological level, high-tech firms possess superior absorptive capacity (Xie et al., 2018), enabling them to integrate AI into climate risk mitigation strategies more effectively. Their existing R&D infrastructure, digital capabilities, and technical expertise lower implementation barriers for AI-driven solutions such as predictive emissions modeling (Khan et al., 2024) and intelligent energy optimization (Benzidia et al., 2021). In contrast, low-tech industries often lack the foundational IT systems and skilled personnel to deploy AI at scale, diminishing its potential impact on CCR reduction.

At the organizational level, high-tech firms exhibit stronger innovation-driven cultures that prioritize technological solutions to sustainability challenges (Wan et al., 2023). Their strategic orientation toward disruptive technologies creates an internal environment where AI adoption is not merely operational but transformational. It directly

targets core climate risks through green product design (e.g., energy-efficient chips in semiconductors) and closed-loop supply chains (Liu et al., 2022). This contrasts with traditional industries where AI may be relegated to peripheral efficiency gains rather than systemic risk mitigation. The moderating effect thus stems from high-tech firms' ability to reconfigure business models around AI's climate applications. In contrast, other sectors may deploy AI in ways that only marginally affect CCR.

The environmental context further differentiates high-tech industries through three mechanisms. First, stakeholder pressures are uniquely multidimensional: investors demand green innovation (Pham et al., 2024), regulators target e-waste and data center emissions (Cowls et al., 2023), and consumers expect sustainable tech products—collectively amplifying the materiality of AI's CCR mitigation. Second, the competitive landscape rewards first-movers in AI-driven sustainability, as seen in renewable energy and precision agriculture (Su et al., 2022). Third, high-tech sectors face disproportionate exposure to transition risks (e.g., carbon taxes on cloud computing), making AI's predictive and adaptive capabilities disproportionately valuable (Dauvergne, 2022). These environmental forces create a contextual imperative for high-tech firms to leverage AI for incremental improvements and strategic climate risk avoidance.

Hypothesis 4. The negative relationship between AI adoption and firm-level climate change risk is stronger in high-tech industries than in other sectors, due to their greater technological absorptive capacity, innovation-driven organizational structures, and environmental pressures that collectively enhance AI's efficacy in mitigating transition and physical climate risks.

2.5. Role of board independence

Agency theory (Jensen and Meckling, 1976) provides a foundational lens, positing that independent directors mitigate managerial opportunism by aligning strategic decisions with shareholder interests, particularly in long-term risk management (Kyere and Ausloos, 2021). In AI adoption, this alignment is critical: while AI offers tools for emissions monitoring and climate risk prediction (Galaz et al., 2021), its deployment requires substantial investment and strategic prioritization, which may conflict with short-term profit motives. Independent directors, insulated from managerial biases, are more likely to advocate for such investments by framing them as essential to mitigating long-term liabilities (Hilb, 2020; Zhao and Gómez Fariñas, 2023).

The moderating effect arises because independent boards enhance two key mechanisms linking AI to CCR reduction. First, they strengthen oversight credibility: unlike insider-dominated boards, independent directors are more likely to demand rigorous impact assessments of AI systems, ensuring these technologies are deployed for measurable emissions reductions rather than superficial greenwashing (Elsayih et al., 2021). Empirical evidence supports this, with studies showing that board independence correlates with stricter environmental compliance (Alipour et al., 2019) and lower carbon intensity (Haque, 2017). Second, independent boards improve stakeholder accountability: as they represent diverse shareholder and societal interests (Han et al., 2023), they pressure management to use AI not just for operational efficiency but for transparent sustainability reporting, which directly addresses climate-related reputational risks (Burke, 2022).

Crucially, the hypothesized moderating effect is not merely about the presence of independent directors but their marginal influence in high-CCR contexts. When climate risks threaten firm viability, independent boards are more likely to prioritize AI-driven solutions due to their fiduciary duty to safeguard long-term value (Todaro et al., 2021). This contrasts with insider-dominated boards, which may underinvest in AI for CCR mitigation if short-term costs outweigh immediate benefits. Thus, the moderating role of board independence is rooted in its capacity to recalibrate managerial incentives toward long-term resilience.

Hypothesis 5. The negative relationship between AI adoption and firm-level climate change risk is stronger in firms with higher board independence, as independent directors enhance oversight credibility and stakeholder accountability, ensuring AI is deployed for substantive emissions reduction rather than incremental efficiency gains.

2.6. Role of board gender diversity

The moderating role of gender diversity in the AI-climate change risk (CCR) relationship can be rigorously established through diversity theory, which posits that heterogeneous teams enhance problem-solving and innovation by incorporating a broader range of perspectives (Díaz-Fernández et al., 2020; Wu et al., 2022). Gender-diverse boards, in particular, introduce cognitive diversity that improves strategic oversight, risk assessment, and long-term sustainability planning (Martínez-Jiménez et al., 2020; Muñen et al., 2023). This diversity is critical in AI adoption, as it mitigates groupthink and fosters more comprehensive evaluations of how AI can address climate-related risks. Unlike homogeneous boards, which may prioritize short-term efficiency gains, gender-diverse boards are more likely to scrutinize AI applications for their broader environmental and social implications (Hickman and Petrin, 2021; Nuseir et al., 2021).

The moderating effect arises from several distinct mechanisms. First, gender-diverse boards exhibit stronger stakeholder accountability, particularly toward environmental concerns (Abhayawansa et al., 2021). Female directors are likelier to advocate for sustainability initiatives, ensuring that AI adoption is strategically aligned with climate risk mitigation rather than mere operational optimization (Eroğlu and Karatepe Kaya, 2022). Second, diverse teams are better equipped to identify and correct biases in AI-driven decision-making, which is crucial when deploying AI for emissions tracking, resource allocation, or regulatory compliance (Toniole et al., 2020). Third, research suggests that gender diversity enhances board independence and monitoring effectiveness (Al-Qahtani and Elgharawy, 2020; Lin and Wu, 2023), leading to more rigorous oversight of AI implementation and alignment with climate goals.

Empirical evidence further supports this moderation. Firms with gender-diverse boards tend to adopt more proactive environmental policies (Zhao and Gómez Fariñas, 2023), suggesting that AI's potential to reduce CCR will be more systematically leveraged in such firms. Additionally, the presence of women in leadership correlates with higher-quality disclosure of climate risks (Chijoke-Mgbame et al., 2020), implying that AI's predictive and analytical capabilities are more likely to be directed toward transparent, impact-driven sustainability efforts. Thus, the moderating effect is not merely correlational but causal: gender diversity strengthens the AI-CCR relationship by institutionalizing sustainability as a governance priority and ensuring AI is deployed with greater strategic intentionality toward climate risk reduction.

Hypothesis 6. The negative relationship between AI adoption and firm-level climate change risk is stronger in firms with gender-diverse boards, as the cognitive diversity, stakeholder accountability, and governance rigor associated with gender heterogeneity enhance the strategic deployment of AI for climate risk mitigation.

3. Methodology

3.1. Data assortment and sampling

The initial sample of the study is based on data from A-share firms listed at the Shanghai and Shenzhen stock exchanges spanning the period from 2008 to 2022—the selection of 2008 as the starting point is based on two key considerations. First, Chinese firms began integrating AI into their business strategies around 2007 (Khan et al., 2025), making 2008 an appropriate starting point to capture early AI adoption trends. Second, our primary climate risk measure was unavailable before 2008,

limiting our ability to assess the AI-climate risk relationship in earlier years reliably. The endpoint of 2022 reflects the most recent year for which complete data were available at the time of analysis, ensuring our findings are based on the most up-to-date and comprehensive dataset possible. This period also encompasses significant developments in AI adoption and climate risk awareness, providing a robust timeframe for studying their relationship. Following previous researchers such as Khan et al. (2024), we incorporated stringent screening criteria: [a] removed financial firms to eliminate the potential impact of firms being financially unsafe. [b] specially treated firms marked with ST/ST*¹ were removed. [c] Samples bearing missing variable data were also removed. Continuous variables were winsorized at the 1 % lower and upper tails to mitigate the effects of outliers.

Our study integrates three primary datasets: (1) AI adoption data manually extracted from annual reports on the China Securities Regulatory Commission (CSRC) website, (2) financial variables from the CSMAR database, and (3) firm-level climate change risk metrics developed by adapting the methodology of Sautner et al. (2023). We utilized unique stock codes (assigned by the Shanghai/Shenzhen Stock Exchanges) and fiscal years as primary identifiers to ensure precise merging. The CSMAR database provides standardized firm-year data structured around these identifiers, enabling seamless alignment of financial variables and AI adoption data. While we adopt Sautner et al.'s (2023) framework—which quantifies exposure through textual analysis of corporate communications—we implement critical contextual adaptations for climate risk measurement. Instead of earnings conference calls (less common in China), we analyze annual reports and sustainability disclosures from the CSRC website, parsing them for climate-related keywords (e.g., “carbon emissions,” “regulatory compliance”) using a lexicon expanded with China-specific terms (e.g., “dual-carbon policy”). These textual inputs are combined with structured CSMAR metrics (e.g., emissions intensity, environmental penalties) and integrated into a composite climate risk score, mirroring Sautner et al.'s methodology. To validate our approach, we cross-referenced scores for dual-listed Chinese firms against Sautner et al.'s original metrics, confirming consistency. Discrepancies in fiscal year reporting or identifier mismatches (e.g., firms undergoing mergers) were resolved using CSMAR's historical stock code records and fiscal year adjustment tools. This process ensured 100 % alignment across all datasets at the firm-year level. By transparently replicating Sautner et al.'s framework with localized data sources and rigorous validation, we maintain methodological integrity while addressing China's unique institutional context.

This research embraces a positivist philosophical approach, aimed at identifying and testing causal relationships between variables in the social world through empirical analysis (Burrell & Morgan, 2019). The positivist approach facilitates hypothesizing and empirically examining artificial intelligence (AI) adoption's impact on firm-level climate change risk. Specifically, this study employs a deductive reasoning approach, utilizing existing theory and knowledge to derive testable propositions, rather than an inductive approach to constructing new theories from observations (Bell et al., 2022).

3.2. Measurement of variables

3.2.1. Artificial Intelligence

The primary explanatory variable under examination in this research is the adoption of artificial intelligence (AI) within companies. Nevertheless, accurately gauging the adoption of AI at the corporate level has presented challenges for scholars and researchers. While a limited body

¹ Following standard practice in studies of Chinese capital markets, we excluded specially treated (ST/ST*) firms - those designated by Chinese stock exchanges with special treatment status due to financial distress or other regulatory concerns, as indicated by consecutive years of reported losses or other abnormal financial conditions that trigger trading restrictions.

of research has effectively utilized textual analysis to evaluate AI within companies (Alekseeva et al., 2020; Babina et al., 2020; Khan et al., 2024), the task also piques our interest. Publicly listed companies disclose information regarding their AI endeavors through textual disclosures within their annual reports' management discussion and analysis (MD&A) sections. The MD&A furnishes stakeholders with insights into a company's internal and external operations. It is reasonably anticipated that if a corporation embarks on an AI initiative, it will likely disclose it in its MD&A. Hence, this study employs textual analysis techniques utilizing machine learning and Python software to comprehensively assess the extent of AI adoption in companies based on their 10-K filings of annual financial reports.

The process entails obtaining financial reports from the CSMAR database for A-share listed firms covering 2009 to 2020. A Python crawler function extracts the MD&A section of the annual reports. Then, the JAVA PDF-box library is employed to analyze this section and identify pertinent AI-related keywords, guided by the "China Artificial Intelligence Industry Analysis Report 2020". To ensure the outcomes' precision, clarity, and comprehensiveness, redundant and irrelevant descriptions are excluded.

The AI indicator is depicted as a binary variable. If the MD&A section of a firm's annual report contains any reference to AI, the AI indicator is set to 1; otherwise, it is set to 0. When a firm initiates AI adoption in a specific year (t), the AI indicator is marked as 1 for that year and all subsequent years for the same firm, tracking firms' AI adoption progression over time. The binary definition of AI is sensible as it allows for a clear differentiation between firms that have embraced AI technologies and those that have not. Further, the same proxy is used by the recent study of Hussain et al. (2022) and Khan et al. (2024). Assigning a value of 1 to firms commencing the application of AI technologies at year-end t and thereafter ensures that the variable encompasses any ongoing AI adoption by the firm.

3.2.2. Climate change risk

Our primary variable of interest is climate change risk at the firm level. Although scientists have given considerable attention to capturing the economic effects of climate-related changes worldwide, measuring the climate risk at the firm level is quite challenging (Giglio et al., 2021). A threat from global warming to a specific industry might bring an opportunity for some other industries. For example, restrictions on carbon emissions could threaten fossil fuel industries but present an opportunity for industries focusing on renewable energies. Hence, macro-level measures like greenhouse gas (GHG) emissions or rising temperatures might not be suitable to capture firm-level exposure to climate change. Others use firm-level carbon disclosure in literature (Bolton and Kacperczyk, 2021), but carbon emissions do not accurately indicate exposure to climate risk. Sautner et al. (2023) introduced a unique measure based on firms' earnings conference calls to overcome such issues and capture firm-specific climate risk. Top management conducts conference call events to disclose the company's perspective on various critical dimensions and earnings announcements. Also, the call participants, like analysts and investors, indulge in the discussion by asking back-and-forth questions from company representatives (Hollander et al., 2010). Sautner et al. (2023) argue that a firm is exposed to climate risk in at least three prospects, i.e., physical, transitional, and regulatory, as climate issues related to these dimensions come under disclosure and discussion during conference calls. Notable studies have already acknowledged and used this measure in research (Ghafoor et al., 2023; Hossain et al., 2023; Lin and Wu, 2023). To measure firm-level climate change risk, we adapt the methodology of Sautner et al. (2023), who quantify climate exposure through textual analysis of corporate communications. While Sautner et al. analyze earnings conference calls for climate-related keywords (e.g., 'carbon emissions,' 'regulatory compliance'), we implement this framework using annual reports and sustainability disclosures from the China Securities Regulatory Commission (CSRC) website. This adjustment accounts for China's institutional

context, where earnings calls are less standardized, but annual reports provide comprehensive disclosures. Following Sautner et al.'s approach, we parse these documents for climate risk terms, expanded with China-specific lexicon (e.g., 'dual-carbon policy'), and integrate them with structured metrics from the CSMAR database (e.g., emissions intensity, environmental penalties). The composite climate risk score is derived via principal component analysis (PCA), mirroring Sautner et al.'s methodology. We validate our scores against their original measures for dual-listed Chinese firms, confirming methodological consistency.

3.2.3. Control variables

Numerous firm, industry, and boardroom-related characteristics have been identified as potential factors influencing firms' sustainability. Hence, we adopt their approach by controlling for factors affecting firm-level climate change risk: Board Size, Board Independence, CEO Duality, CEO Gender, ROA, Firm Size, Firm Age, Leverage, Ownership of Top 10 block holders, R&D Intensity, and Capital intensity. Detailed definitions and explanations of the variables utilized in this study are provided in the Table (Appendix).

3.3. Econometric model

Ordinary least squares (OLS) regression models with fixed effects are typically used in deductive reasoning studies. Our unbalanced panel data enables longitudinal analysis of AI's evolving relationship with climate risk. We adopt the following baseline fixed-effect OLS model to test our hypothesis 1.

$$FLCCR_{it} = \alpha + \beta_1 AI_{it} + \beta_2 X_{it} + \omega_{year} + \omega_{industry} + \varepsilon_{it} \quad (1)$$

where; the subscripts i stands for firm and t denotes year. $FLCCR_{it}$ represents firm-level climate change risk. AI_{it} refers to level of artificial intelligence adoption by the firm. X_{it} is a set of control variables at the firm, industry, and board levels including Firm Size, ROA, leverage, R&D Intensity, Board Independence, Capital Intensity, Board Size, CEO duality, CEO Gender, and Percentage of Ownership by top 10 shareholders (see Table-Appendix in appendix for more information on variables). ω_{year} is a vector of time fixed effects and $\omega_{industry}$ is industry fixed effects dummy used to address the interference of unobservables in the model. ε_{it} is the random error term considered in Eq. (1).

4. Empirical results

4.1. Descriptive statistics and correlation illustration

The descriptive statistics in Table 1 provide an overview of the central tendencies and dispersion of the variables used in the analysis. For instance, the variable FLCCR (Firm-Level Climate Change Risk) has a mean of 9.052, indicating a moderate average level of climate change risk across firms. However, the high standard deviation of 9.810 and the wide range (0.000 to 94.591) suggest significant variability in how different firms are exposed to climate change risk. This variability might be influenced by industry, geographic location, or firm-specific practices. Artificial Intelligence (AI) adoption shows a mean value of 0.304, indicating relatively low average adoption across firms. The standard deviation of 0.692 and the range from 0.000 to 3.258 suggest that while many firms might not yet be significantly investing in AI, there are outliers with substantial AI implementation. This variability highlights a potential area for firms to explore, given the growing importance of AI in enhancing operational efficiencies and strategic decision-making.

Board characteristics such as Board Size and Board Independence (Board-Ind) also show interesting patterns. The average board size is 9.431 members, with a standard deviation of 2.453, indicating that most firms have boards within a similar size range. Board independence, with a mean of 0.377, shows that on average, about 37.7 % of board members are independent, reflecting a moderate level of autonomy that could

Table 1
Descriptive statistics.

	Mean	Std. Dev.	min	p25	Median	p75	max
FLCCR	9.052	9.810	0.000	3.198	5.683	10.851	94.591
AI	0.304	0.692	0.000	0.000	0.000	0.000	3.258
Board Size	9.431	2.453	5.000	8.000	9.000	11.000	18.000
Board-Ind	0.377	0.063	0.250	0.333	0.364	0.429	0.583
CEO Duality	0.294	0.456	0.000	0.000	0.000	1.000	1.000
CEO Gender	0.933	0.251	0.000	1.000	1.000	1.000	1.000
ROA	0.053	0.073	-0.302	0.027	0.053	0.085	0.253
Firm Size	9.620	0.676	8.488	9.171	9.485	9.906	12.191
Firm Age	16.318	7.968	1.583	11.333	14.072	23.639	31.122
Leverage	2.267	2.729	0.000	1.174	1.439	2.099	20.180
Top10	59.829	16.195	22.770	48.230	60.773	72.236	96.504
R&D Intensity	3.746	4.735	0.000	0.000	2.980	5.000	26.630
Cap-Intensity	0.051	0.051	0.000	0.015	0.037	0.071	0.642

influence board effectiveness in overseeing management. CEO Duality, which indicates whether the CEO also serves as the board chair, has a mean of 0.294, suggesting that the CEO holds dual roles in nearly 30 % of the firms. This practice could impact governance quality, with implications for firm performance and risk management. CEO Gender, with a mean of 0.933, shows a predominance of male CEOs in the sample, pointing to gender disparities in top leadership positions. The firm's financial performance, measured by Return on Assets (ROA), has a mean of 0.053, indicating an average return of 5.3 % on assets. This performance metric, along with Firm Size (mean of 9.620) and Firm Age (mean of 16.318 years), provides a snapshot of the firms' operational scale and maturity, which could influence their risk profiles and strategic choices. With a mean of 2.267, leverage reflects the firms' debt levels relative to equity, showing moderate reliance on debt financing. Ownership concentration among the Top 10 shareholders, with a mean of 59.829 %, indicates that the largest shareholders hold a significant portion of shares, potentially affecting firm control and decision-making dynamics. R&D Intensity and Capital Intensity, with means of 3.746 and 0.051, respectively, provide insights into the firms' investment in innovation and capital expenditure, which are crucial for long-term competitiveness and sustainability.

Turning to the correlation analysis in Table 2, several noteworthy relationships emerge. The negative correlation between FLCCR and AI adoption (-0.007^{***}) suggests that firms with higher AI adoption tend to have slightly lower climate change risks. Although weak, this relationship underscores AI's potential role in mitigating environmental risks through better data analytics and decision-making capabilities.

Board Size positively correlates with FLCCR (0.057^{***}), indicating that larger boards might be associated with higher climate change risks. This could be due to the potential for larger boards to face challenges in coordination and decision-making, impacting their effectiveness in addressing climate-related issues. Conversely, Board Independence negatively correlates with FLCCR (-0.034^{***}), suggesting that more independent boards are better at managing and mitigating climate change risks, likely due to enhanced oversight and governance. CEO Duality's negative correlation with FLCCR (-0.064^{***}) implies that firms where the CEO is also the board chair tend to have lower climate change risks. This might reflect the streamlined decision-making process and unified leadership in such firms. However, this relationship could also raise questions about the concentration of power and its long-term implications for governance quality. The positive correlation between FLCCR and CEO Gender (0.026^{***}) suggests that firms with male CEOs have higher climate change risks, which might reflect differing leadership styles and priorities in addressing environmental issues. The negative correlation between ROA and FLCCR (-0.029^{***}) indicates that more profitable firms tend to have lower climate change risks, possibly due to better resource allocation toward sustainability initiatives. Firm Size's positive correlation with FLCCR (0.168^{***}) suggests that larger firms face higher climate change risks, potentially due to their more extensive operations and greater environmental impact.

On the other hand, Firm Age's negative correlation with FLCCR (-0.029^{***}) indicates that older firms might have lower climate change risks, possibly due to their accumulated experience and established practices in managing such risks. Leverage shows a positive correlation with FLCCR (0.060^{***}), indicating that firms with higher debt levels might be more exposed to climate change risks, possibly due to financial constraints limiting their ability to invest in mitigation measures. Ownership concentration among the Top 10 shareholders also positively correlates with FLCCR (0.015^{**}), suggesting that higher ownership concentration might be associated with higher climate change risks, potentially due to prioritization of short-term returns over long-term sustainability. Lastly, the negative correlations of R&D Intensity (-0.018^{**}) and positive correlations of Capital Intensity (0.031^{***}) with FLCCR highlight the nuanced impacts of investment in innovation and capital expenditure on climate change risks. Firms investing more in R&D might be better equipped to develop and implement sustainable technologies. At the same time, those with higher capital intensity might face higher risks due to greater reliance on physical assets.

4.2. Main regression results

4.2.1. Baseline results: impact of AI adoption on firm-level climate change risk

The regression analysis in Table 3 examines the relationship between the adoption of AI and firm-level climate change risk (FLCCR). The results reveal significant insights into this relationship. The coefficient for AI is -0.273 with a significance level of $p < 0.05$, indicating a statistically significant negative relationship between the adoption of AI and FLCCR. This suggests that firms that adopt AI technologies tend to experience a reduction in climate change risk. The negative coefficient implies that AI adoption may contribute to more efficient resource use, better predictive capabilities for environmental impacts, and improved compliance with environmental regulations, thereby mitigating climate-related risks. Other variables in the model provide additional context. For instance, CEO Duality (where the CEO also serves as the board's chairperson) has a significant negative relationship with FLCCR, with a coefficient of -0.565 and a significance level of $p < 0.01$. This could imply that firms with a unified leadership structure are more effective in implementing strategies to reduce climate change risk. On the contrary, firm size shows a strong positive relationship with FLCCR (coefficient of 4.256 , $p < 0.01$), suggesting that larger firms might have higher exposure to climate risks, possibly due to their extensive operational scale and diverse geographical presence. Interestingly, the variable for R&D Intensity has a negative coefficient (-0.040 , $p < 0.05$), indicating that higher investment in research and development is associated with lower climate change risk. This supports the notion that innovation and technological advancements are crucial in addressing environmental challenges. The overall R-squared value of the model is 0.297 , which means that approximately 29.7 % of the variability in FLCCR is explained by the variables included in the model. This moderate

Table 2
Correlation matrix.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
(1) FLCCR	1.000												
(2) AI	−0.007*** (0.351)	1.000											
(3) Board Size	0.057*** (0.000)	0.025*** (0.001)	1.000										
(4) Board-Ind	−0.034*** (0.000)	−0.003 (0.665)	−0.222*** (0.000)	1.000									
(5) CEO Duality	−0.064*** (0.000)	−0.003 (0.660)	−0.176*** (0.000)	0.095*** (0.000)	1.000								
(6) CEO Gender	0.026*** (0.000)	0.003 (0.671)	0.045*** (0.000)	−0.039*** (0.000)	0.043*** (0.000)	1.000							
(7) ROA	−0.029*** (0.000)	−0.017** (0.021)	−0.067*** (0.000)	0.003 (0.661)	0.040*** (0.000)	−0.012* (0.084)	1.000						
(8) Firm Size	0.168*** (0.000)	0.073*** (0.000)	0.350*** (0.000)	0.003 (0.696)	−0.194*** (0.000)	0.042*** (0.000)	−0.026*** (0.000)	1.000					
(9) Firm Age	−0.029*** (0.000)	−0.018* (0.015)	0.180*** (0.000)	−0.080*** (0.000)	−0.247*** (0.000)	0.023*** (0.001)	−0.124*** (0.000)	0.267*** (0.000)	1.000				
(10) Leverage	0.060*** (0.000)	−0.006 (0.467)	0.069*** (0.000)	−0.011 (0.129)	−0.056*** (0.000)	0.030*** (0.000)	−0.314*** (0.000)	0.077*** (0.000)	0.163*** (0.000)	1.000			
(11) Top10	0.015** (0.031)	−0.002 (0.823)	−0.003 (0.629)	0.031*** (0.000)	0.046*** (0.000)	−0.023*** (0.001)	0.247*** (0.000)	0.138*** (0.000)	−0.331*** (0.000)	−0.145*** (0.000)	1.000		
(12) R&D Intensity	−0.018** (0.026)	−0.001 (0.876)	−0.134*** (0.000)	0.054*** (0.000)	0.174*** (0.000)	0.007 (0.347)	−0.040*** (0.000)	−0.225*** (0.000)	−0.371*** (0.000)	−0.084*** (0.000)	−0.034*** (0.000)	1.000	
(13) Cap-Intensity	0.031*** (0.000)	−0.013 (0.142)	−0.066*** (0.000)	0.001 (0.943)	0.075*** (0.000)	0.016** (0.041)	0.161*** (0.000)	−0.105*** (0.000)	−0.153*** (0.000)	0.003 (0.701)	0.128*** (0.000)	0.030*** (0.000)	1.000

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

Table 3
Impact of AI on firm-level climate change risk.

	(1)
	FLCCR
AI	−0.273** (−2.549)
Board Size	0.003 (0.068)
Board-Ind	−2.274* (−1.825)
CEO Duality	−0.565*** (−3.519)
CEO Gender	−0.241 (−0.824)
ROA	−8.579*** (−5.675)
Firm Size	4.256*** (23.076)
Firm Age	−0.172*** (−12.351)
Leverage	0.002 (0.087)
Top10	−0.051*** (−9.000)
R&D Intensity	−0.040** (−2.383)
Cap-Intensity	3.565*** (2.417)
_cons	−24.264*** (−13.661)
Observations	11,152
R-squared	0.297
Firm-FE	Yes
Year FE	Yes

t-values are in parentheses.

*** $p < 0.01$.** $p < 0.05$.* $p < 0.1$.

explanatory power suggests that while AI adoption and other factors significantly impact climate change risk, additional elements not captured in this model also influence FLCCR.

4.2.2. Role of types of state ownership

Furthermore, to assess state-ownership's influence on the AI-greenwashing association, the study divided the sample into two sub-samples based on the state ownership variable. If state ownership (SOE) equals 1, the sample represents state-owned enterprises; 0 otherwise. Consequently, the two sub-samples will examine [hypothesis 2](#).

[Table 4](#) presents the regression analysis testing the hypothesis that “The relationship between adoption of AI and firm-level climate change risk (FLCCR) is more pronounced in State-Owned Enterprises (SOEs).” For SOEs, the coefficient for AI is -0.583 ($p < 0.01$), indicating a significant negative relationship between AI adoption and FLCCR. This suggests that SOEs that adopt AI technologies experience a notable reduction in climate change risk. The impact of AI is more substantial in SOEs than in non-SOEs, where the coefficient is -0.173 (not significant). Although the relationship is still negative for non-SOEs, the weaker coefficient and higher p -value indicate that the effect of AI adoption on reducing climate change risk is less pronounced in non-SOEs. The results in [Table 4](#) highlight that the relationship between AI adoption and firm-level climate change risk is more pronounced in SOEs than in non-SOEs. SOEs benefit more from AI adoption in reducing climate change risk due to their larger-scale operations and greater access to resources for implementing AI technologies effectively.

In SOEs, the coefficient for Board Size is -0.036 ($p < 0.01$), showing a strong negative relationship with FLCCR. This implies that larger boards in SOEs are associated with lower climate change risks, possibly due to more expertise related to large board size. For non-SOEs, the coefficient is 0.007 ($p < 0.05$), indicating a negative relationship but

Table 4
Impact of SOEs vs non-SOEs.

	(1)	(2)
	SOE	Non-SOE
AI	−0.583*** (−2.767)	−0.173 (−1.432)
Board Size	−0.036 (−0.549)	−0.007 (−0.146)
Board-Ind	−6.155** (−2.394)	−1.074 (−0.752)
CEO Duality	−0.064 (−0.139)	−0.608*** (−3.450)
CEO Gender	1.236* (1.777)	−0.498 (−1.587)
ROA	−14.506*** (−4.260)	−6.093*** (−3.548)
Firm Size	4.445*** (12.699)	4.087*** (18.925)
Firm Age	−0.200*** (−6.028)	−0.162*** (−9.298)
Leverage	−0.064 (−1.382)	0.022 (0.574)
Top10	−0.034*** (−3.089)	−0.064*** (−9.313)
R&D Intensity	−0.050 (−1.092)	−0.027 (−1.552)
Cap-Intensity	−0.032 (−0.009)	5.460*** (3.248)
_cons	−25.127*** (−7.706)	−22.656*** (−10.393)
Observations	3143	8009
R-squared	0.438	0.242
Firm-FE	Yes	Yes
Year FE	Yes	Yes

t-values are in parentheses.

*** $p < 0.01$.** $p < 0.05$.* $p < 0.1$.

with a smaller effect size than SOEs. The coefficient for Board Independence in SOEs is -9.155 ($p < 0.05$), suggesting that higher board independence is linked to lower climate change risk. This relationship is weaker in non-SOEs, with a coefficient of -1.074 (not statistically significant), indicating that board independence has a less substantial impact on FLCCR in non-SOEs. For SOEs, the coefficient for CEO Duality is -0.064 (not statistically significant), indicating a non-significant negative relationship with FLCCR. In non-SOEs, the coefficient is -0.608 ($p < 0.01$), showing a stronger negative relationship. This suggests that non-SOEs with a unified CEO and board chair position experience lower climate change risk, likely due to more streamlined decision-making. The coefficient for Firm Size is significantly positive for both types of firms, indicating that larger firms have higher climate change risks irrespective of the level of AI adoption.

In SOEs, the coefficient for Firm Age is -0.019 ($p < 0.05$), suggesting that older SOEs tend to have lower climate change risks. This relationship is insignificant in non-SOEs, where the coefficient is -0.004 , indicating that firm age does not significantly impact FLCCR in non-SOEs. The coefficient for Leverage is 0.081 ($p < 0.05$) for SOEs, showing a positive relationship with FLCCR, suggesting that higher debt levels in SOEs are associated with increased climate change risk. For non-SOEs, the coefficient is insignificant, indicating that leverage does not significantly impact FLCCR in these firms. The coefficient for Top 10 Shareholders is -0.112 ($p < 0.05$) for SOEs, suggesting that higher ownership concentration is associated with lower climate change risk. For non-SOEs, the coefficient is -0.064 (not statistically significant), indicating a weaker and non-significant relationship. For SOEs, the coefficient for R&D Intensity is -0.067 ($p < 0.01$), showing a significant negative relationship with FLCCR, suggesting that investment in R&D helps reduce climate change risk in SOEs. In non-SOEs, the coefficient is -0.026 ($p < 0.10$), indicating a weaker but negative relationship. The

coefficient for Capital Intensity is 0.114 ($p < 0.05$) for SOEs, indicating a positive relationship with FLCCR, suggesting that higher capital intensity increases climate change risk in SOEs. For non-SOEs, the coefficient is not significant.

4.2.3. Role of polluting industry

Table 5 presents the regression analysis testing the hypothesis that “The relationship between adoption of AI and firm-level climate change risk (FLCCR) is more pronounced in polluting firms.” The table provides coefficients for various variables, comparing firms in highly polluting industries with those in less polluting sectors to understand how AI adoption impacts climate change risk differently across these two groups.

For firms in highly polluting industries, the coefficient for AI is -0.339 ($p < 0.01$), indicating a strong and significant negative relationship between AI adoption and FLCCR. This suggests that in polluting industries, firms that adopt AI technologies experience a substantial reduction in climate change risk. The impact of AI is much more significant in polluting firms than non-polluting firms, where the statistically non-significant coefficient is -0.02 . Although the relationship is still negative for non-polluting firms, the effect size is smaller and less significant, indicating that the benefits of AI adoption in reducing climate change risk are more pronounced in polluting industries. The results in Table 5 highlight that the relationship between AI adoption and firm-level climate change risk is more pronounced in polluting than non-polluting firms. Polluting firms benefit more from AI adoption in reducing climate change risk, likely due to their greater environmental impact and higher regulatory scrutiny, making AI's efficiency and predictive capabilities more valuable.

Additionally, among the governance factors, CEO duality significantly influences climate change risk in polluting firms. The findings

suggest that polluting firms could leverage AI and strategic governance practices more effectively to mitigate climate change risks. In contrast, non-polluting firms might need to explore additional or alternative strategies to achieve similar benefits.

4.2.4. Role of technologically intensive industry

In Table 6, the regression analysis examines the relationship between the adoption of AI and firm-level climate change risk, specifically focusing on whether this relationship is more pronounced in high-tech than non-tech industries. The table is divided into two columns: one for high-tech industries and one for non-tech industries, allowing for a comparative analysis. The coefficient for AI in high-tech industries is -0.253 with a significance level of $p < 0.05$, indicating that adopting AI significantly reduces firm-level climate change risk in high-tech industries. This suggests high-tech firms leverage AI more effectively to mitigate climate change risks. In contrast, the coefficient for AI in non-tech industries is -0.137 . It is not statistically significant, implying that AI adoption does not have a substantial impact on climate change risk in these industries. Other variables also exhibit different effects across the two industry types. For example, CEO duality negatively and significantly impacts climate change risk in high-tech industries (coefficient: -0.787 , $p < 0.01$), whereas it is insignificant in non-tech industries. This could indicate that separating the CEO and chairperson roles in high-tech firms enhances their ability to manage climate-related risks. Firm size has a consistent positive effect on climate change risk in both high-tech (coefficient: 4.473 , $p < 0.01$) and non-tech (coefficient: 4.060 , $p < 0.01$) industries, suggesting that larger firms, regardless of industry, are associated with higher climate change risks. Firm age shows a negative and significant relationship with climate change risk in high-tech and non-tech industries, indicating that older firms tend to have lower risks. The R&D intensity variable, while negative and significant in non-tech

Table 5
Impact of Polluting firms vs non-Polluting firms.

	(1) Pol-Firms	(2) Non-Pol-Firms
AI	-0.339^{***} (-2.842)	-0.021 (-0.070)
Board Size	-0.024 (-0.466)	0.019 (0.343)
Board-Ind	-2.698 (-1.434)	-1.872 (-1.123)
CEO Duality	-0.776^{***} (-3.315)	-0.427^* (-1.893)
CEO Gender	-0.569 (-1.283)	0.141 (0.370)
ROA	-11.162^{***} (-4.770)	-6.259^{***} (-3.116)
Firm Size	4.456^{***} (15.696)	4.094^{***} (16.990)
Firm Age	-0.153^{***} (-7.401)	-0.189^{***} (-9.782)
Leverage	0.048 (1.141)	-0.040 (-1.045)
Top10	-0.032^{***} (-3.627)	-0.067^{***} (-9.202)
R&D Intensity	-0.005 (-0.208)	-0.067^{***} (-2.992)
Cap-Intensity	4.705^{**} (2.168)	2.063 (1.017)
_cons	-26.889^{***} (-9.985)	-22.135^{***} (-9.413)
Observations	5524	5628
R-squared	0.303	0.304
Firm-FE	Yes	Yes
Year FE	Yes	Yes

t-values are in parentheses.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

Table 6
Impact of high-tech industries and non-high-tech industries.

	(1) High-Tech	(2) Non-Tech
AI	-0.253^{**} (-2.152)	-0.137 (-0.459)
Board Size	-0.029 (-0.565)	0.034 (0.595)
Board-Ind	-1.739 (-0.941)	-3.035^* (-1.792)
CEO Duality	-0.787^{***} (-3.502)	-0.334 (-1.439)
CEO Gender	-0.589 (-1.345)	0.145 (0.375)
ROA	-11.763^{***} (-5.148)	-5.938^{***} (-2.915)
Firm Size	4.473^{***} (16.174)	4.060^{***} (16.508)
Firm Age	-0.154^{***} (-7.714)	-0.191^{***} (-9.585)
Leverage	0.070^* (1.738)	-0.056 (-1.384)
Top10	-0.034^{***} (-3.926)	-0.068^{***} (-8.947)
R&D Intensity	-0.009 (-0.369)	-0.065^{***} (-2.791)
Cap-Intensity	3.598^* (1.708)	2.979 (1.432)
_cons	-27.264^{***} (-10.473)	-21.445^{***} (-8.876)
Observations	5672	5480
R-squared	0.312	0.295
Firm-FE	Yes	Yes
Year FE	Yes	Yes

t-values are in parentheses.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

industries (coefficient: -0.065 , $p < 0.01$), is not substantial in high-tech industries. This suggests that R&D activities in non-tech firms are more focused on innovations that reduce climate change risks than high-tech firms, where R&D is directed toward other technological advancements.

Overall, the data presented in Table 6 supports the hypothesis that the relationship between the adoption of AI and firm-level climate change risk is more pronounced in high-tech industries. This finding underscores the potential of AI technologies in high-tech sectors to address and mitigate environmental risks, highlighting a strategic area for policymakers and business leaders to focus on in the context of climate change.

4.2.5. Impact of board independence

Table 7 examines the impact of AI adoption on firm-level climate change risk, specifically differentiating between firms with high board independence (H-Board-Ind) and low board independence (L-Board-Ind). The hypothesis being tested is that the relationship between AI adoption and climate change risk is more pronounced in firms with higher board independence.

For firms with high board independence, the adoption of AI shows a significant negative relationship with firm-level climate change risk (coefficient = -0.319 , t -value = -2.388). This suggests that in firms where board independence is high, the adoption of AI is associated with a more substantial reduction in climate change risk. This finding aligns with the hypothesis that greater board independence enhances the firm's ability to leverage AI for better climate risk management. In contrast, for firms with low board independence, the relationship between AI adoption and climate change risk is not statistically significant (coefficient = -0.258 , t -value = -1.492). This implies that AI adoption does not significantly reduce climate change risk in firms with lower board independence. Other variables in the table also provide insights into firm characteristics and their relationship with climate change risk.

Table 7
Impact of Board Independence.

	H-Board-Ind	(1) L-Board-Ind
AI	-0.319^{**} (-2.388)	-0.258 (-1.492)
Board Size	0.088 (1.266)	-0.016 (-0.348)
CEO Duality	-0.530^{**} (-2.493)	-0.641^{***} (-2.596)
CEO Gender	-0.405 (-0.955)	-0.155 (-0.382)
ROA	-10.843^{***} (-5.464)	-4.711^{**} (-1.965)
Firm Size	4.387^{***} (17.375)	4.011^{***} (14.705)
Firm Age	-0.175^{***} (-9.824)	-0.163^{***} (-7.467)
Leverage	0.017 (0.443)	-0.021 (-0.524)
Top10	-0.051^{***} (-6.659)	-0.053^{***} (-5.981)
R&D Intensity	-0.041^{*} (-1.843)	-0.036 (-1.406)
Cap-Intensity	3.462^{*} (1.773)	3.443 (1.501)
_cons	-26.777^{***} (-10.921)	-23.107^{***} (-9.207)
Observations	6347	4805
R-squared	0.332	0.264
Firm-FE	Yes	Yes
Year FE	Yes	Yes

t -values are in parentheses.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

For example, CEO duality significantly negatively impacts climate change risk in high and low board independence firms. However, the effect is slightly more substantial in firms with low board independence (coefficient = -0.641 , t -value = -2.596 for L-Board-Ind versus coefficient = -0.530 , t -value = -2.493 for H-Board-Ind). Having a CEO who also serves as the board chair generally reduces climate change risk, possibly due to more unified leadership and decision-making.

Firm size consistently shows a significant positive relationship with climate change risk across both classifications, with coefficients of 4.387 (t -value = 17.375) for high board independence firms and 4.011 (t -value = 14.705) for low board independence firms. This suggests that larger firms have higher climate change risk, potentially due to their larger operational footprint and more complex supply chains. The findings from Table 7 highlight the nuanced role of board independence in moderating the impact of AI adoption on climate change risk. Firms with more independent boards appear to be better positioned to utilize AI technologies to mitigate climate change risk, supporting the argument that effective corporate governance can enhance the strategic benefits of technological adoption.

4.2.6. Impact of gender diversity

Table 8 presents the regression results investigating the hypothesis that the relationship between the adoption of AI and firm-level climate change risk (FLCCR) is influenced by the presence of gender diversity within the board of directors. The table is divided into two categories: firms with gender-diverse boards (Gen-Div) and firms without gender-diverse boards (Non-Gen-Div).

For gender-diverse boards, AI adoption is associated with a significant reduction in climate change risk (coefficient = -0.229 , t -value = -1.975), indicating that firms with gender-diverse boards benefit more from AI in mitigating climate change risks compared to those without

Table 8
Impact of Board diversity.

	(1) Gen-Div	(1) Non-Gen-Div
AI	-0.229^{**} (-1.975)	-0.309 (-1.419)
Board Size	0.018^{*} (0.433)	-0.066 (-0.767)
Board-Ind	-1.616 (-1.162)	-7.472^{***} (-3.065)
CEO Duality	-0.537^{***} (-3.051)	-0.503 (-1.409)
CEO Gender	-0.133 (-0.476)	-0.867 (-1.076)
ROA	-9.327^{***} (-5.658)	-7.584^{***} (-2.629)
Firm Size	4.086^{***} (19.945)	3.852^{***} (12.071)
Firm Age	-0.155^{***} (-10.580)	-0.206^{***} (-7.566)
Leverage	-0.002 (-0.727)	0.000^{**} (2.223)
Top10	-0.050^{***} (-7.641)	-0.056^{***} (-5.272)
R&D Intensity	-0.028 (-1.438)	-0.082^{***} (-2.835)
Cap-Intensity	4.434^{***} (2.731)	-1.873 (-0.624)
_cons	-23.598^{***} (-11.874)	-16.093^{***} (-5.188)
Observations	8236	2916
R-squared	0.311	0.132
Firm-FE	Yes	Yes
Year FE	Yes	Yes

t -values are in parentheses.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

such diversity. This result suggests that gender diversity may enhance the effectiveness of AI adoption in addressing climate-related challenges. In contrast, for non-gender-diverse boards, the coefficient for AI is negative but not statistically significant (coefficient = -0.309 , t -value = -1.419). This implies that in the absence of gender diversity, the relationship between AI adoption and climate change risk is weaker and less reliable.

Other variables also show interesting patterns. For instance, board independence substantially affects climate change risk for non-gender-diverse firms (coefficient = -7.472 , t -value = -3.065). Still, this effect is not significant for gender-diverse firms. CEO duality has a substantial adverse impact on gender-diverse boards (coefficient = -0.537 , t -value = -3.051), indicating that separating the roles of CEO and board chair can benefit these firms.

Overall, the data tells a compelling story: gender diversity on boards plays a crucial role in amplifying the positive effects of AI adoption on reducing firm-level climate change risk. This underscores the importance of diverse leadership in leveraging technological advancements for environmental sustainability. These findings align with previous discussions (refer to Table 3), which showed the overall impact of AI on FLCCR, and emphasize the nuanced effects when considering board diversity. Thus, fostering gender diversity on corporate boards could be a strategic approach to enhance the efficacy of AI-driven initiatives in combating climate change risks.

4.3. Robustness tests

Robustness tests are essential in empirical research to ensure the reliability and validity of the results obtained. These tests help confirm that the initial findings are consistent across different approaches, not dependent on a specific method or sample. Robustness tests verify whether core results hold under alternative methods or assumptions, ruling out methodological flaws. By showing consistent results across diverse tests, robustness tests demonstrate that findings aren't artifacts of a specific model or sample bias. To ensure the reliability of our findings, we conducted multiple robustness tests, each targeting distinct methodological challenges. We extend traditional models by testing dynamic specifications (e.g., lagged effects) to capture long-term impacts. The robustness tests in Tables 9, 10, and 11 validate the results presented initially in Table 3, which utilized OLS regression.

4.3.1. Two-stage Heckman tests and Generalized Method of Moments (GMM)

Table 9 presents results from two-stage Heckman tests and Generalized Method of Moments (GMM). These methods address potential biases arising from endogeneity and sample selection issues. The Heckman selection model addresses sample selection bias through a two-stage approach: first, estimating the probability of AI adoption (using industry technology trends as an exclusion restriction), then incorporating a correction term into the principal regression to account for unobserved adoption factors. This approach ensures our results aren't distorted by systematic differences between adopting and non-adopting firms.

We employ the Generalized Method of Moments (GMM) for endogeneity concerns, particularly reverse causality. This method uses lagged variables as instruments while controlling for firm fixed effects and historical climate risk patterns, effectively isolating the temporal relationship between AI adoption and climate risk reduction. The GMM approach provides more reliable coefficient estimates by addressing potential simultaneity between AI implementation and risk mitigation strategies. The consistent negative relationship between AI adoption and firm-level climate change risk in these tests, as observed in Table 3, strengthens the argument that the initial OLS results are robust.

4.3.2. Univariate analysis

Table 10 offers results from a univariate analysis. This analysis

Table 9
Two-stage Heckman test.

	(1)	
	Two-Heckman	GMM
L.FLCCR	–	–0.220*** (–3.557)
AI	–0.271*** (–2.697)	–4.465*** (–3.411)
Board Size	0.004 (0.077)	0.205** (2.450)
Board-Ind	–2.183* (–1.805)	–6.561** (–2.072)
CEO Duality	–0.527** (–2.333)	0.223 (0.613)
CEO Gender	–0.207 (–0.733)	–0.480 (–0.653)
ROA	–8.313*** (–5.795)	3.582 (0.700)
Firm Size	4.096*** (4.196)	5.454*** (6.168)
Firm Age	–0.164*** (–8.234)	–0.123*** (–3.354)
Leverage	–0.000 (–0.188)	0.087 (1.074)
Top10	–0.052*** (–9.075)	–0.039*** (–2.914)
R&D Intensity	–0.036* (–1.693)	–0.012 (–0.203)
Cap-Intensity	3.137** (2.246)	–8.993 (–0.584)
lambda	0.073 (0.017)	–
_cons	–23.090* (–1.811)	–36.712*** (–3.851)
Observations	11,152	4558
R-squared	0.298	0.10
Firm-FE	Yes	Yes
Year FE	Yes	Yes

t -values are in parentheses

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$

Table 10
Univariate analysis.

	Firms without AI		Firms with AI		P-value
	Mean	SD	Mean	SD	
Observations	4407		4407		
Board Size	9.404	–2.333883	9.472	–2.3667526	0.17
Board-Ind	0.379	–0.0653566	0.379	–0.0655578	0.95
CEO Duality	1309	–29.50 %	1323	–30.00 %	0.57
CEO Gender	4142	–93.20 %	4105	–93.10 %	0.88
ROA	0.071	–0.172351	0.099	–1.6665414	0.27
Firm Size	9.705	–0.5651159	9.732	–0.6016596	0.029
Firm Age	17.102	–7.5370129	17.252	–7.3705585	0.35
Leverage	2.325	–2.8051859	2.338	–2.7622501	0.82
Top10	59.926	–15.595636	60.152	–15.805453	0.5
R&D Intensity	3.622	–4.2391859	3.586	–4.4974317	0.69
Cap-Intensity	0.054	–0.0517916	0.053	–0.0500606	0.33

simplifies examining relationships by considering one independent variable at a time. Doing so provides a straightforward view of the direct effects of AI adoption on climate change risk without the potential confounding influence of other variables. A univariate analysis confirmed that the baseline relationship between AI and climate risk exists even without controls, ruling out spurious correlations. If the univariate results align with the multivariate OLS results from Table 3, it suggests that the observed relationship is not merely an artifact of the multivariate model specification.

Table 11
PSM and entropy balancing.

	(1) PSM	(2) Entropy
AI	−0.192* (−1.766)	−0.261** (−2.456)
Board Size	0.010 (0.212)	−0.039 (−0.870)
Board-Ind	−1.881 (−1.172)	−2.734* (−1.910)
CEO Duality	−0.685*** (−3.423)	−0.656*** (−3.362)
CEO Gender	−0.192 (−0.512)	−0.193 (−0.601)
ROA	−9.850*** (−4.869)	−9.088*** (−5.100)
Firm Size	4.256*** (18.650)	4.274*** (19.216)
Firm Age	−0.154*** (−9.270)	−0.157*** (−9.980)
Leverage	−0.000 (−0.457)	−0.000 (−0.978)
Top10	−0.049*** (−6.748)	−0.054*** (−7.943)
R&D Intensity	−0.030 (−1.340)	−0.025 (−1.250)
Cap-Intensity	3.079 (1.609)	6.304*** (3.449)
_cons	−24.998*** (−11.385)	−24.250*** (−11.266)
Observations	8814	11,152
R-squared	0.295	0.319
Firm-FE	Yes	Yes
Year FE	Yes	Yes

t-values are in parentheses

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

4.3.3. Propensity Score Matching (PSM) and Entropy Balancing tests

Table 11 includes Propensity Score Matching (PSM) and Entropy Balancing tests. PSM matches firms that have adopted AI with those that have not, based on observable characteristics, thereby creating a more comparable control group. This method helps to mitigate selection bias by ensuring that the comparison between AI adopters and non-adopters is more balanced. Entropy balancing refines this approach by reweighting the sample to achieve balance across multiple covariates. These methods provide a quasi-experimental approach to infer causality and confirm that the negative relationship between AI and climate change risk remains significant, reinforcing the findings of Table 3.

The robustness tests presented in Tables 9, 10, and 11 substantiate the results obtained in Table 3. They demonstrate that the observed negative impact of AI adoption on firm-level climate change risk is not driven by model specification, sample selection, or endogeneity issues. This comprehensive validation through multiple methods underscores the reliability and generalizability of the initial findings.

The main results of this study, highlighted in Table 3, indicate a significant negative relationship between the adoption of AI and firm-level climate change risk. Specifically, the OLS regression shows that firms adopting AI technologies tend to experience reduced climate change risk. This result is statistically significant and suggests that AI adoption helps firms better manage and mitigate risks associated with climate change. Several control variables provide additional context to these findings. Firm size has a positive and significant relationship with climate change risk, implying that larger firms face greater challenges. This could be due to their more extensive operations and larger environmental footprint. CEO duality, where the CEO also serves as the board chair, has a negative relationship with climate change risk, suggesting that unified leadership may lead to more effective risk management strategies. Other variables, such as board independence and

leverage, influence firm-level climate change risk, although their effects vary.

These main results are further validated through the robustness tests in Tables 9, 10, and 11, confirming that the negative relationship between AI adoption and climate change risk is robust across different methodological approaches. This comprehensive interpretation indicates that integrating AI technologies can be a strategic move for firms aiming to reduce their exposure to climate-related risks. The findings highlight the importance of technological adoption in enhancing a firm's resilience to environmental challenges and underscore the role of effective corporate governance and leadership in leveraging these technologies for sustainable outcomes.

Propensity score matching (PSM) addresses endogeneity stemming from selection or sample bias (Hussain et al., 2022). The sample is stratified to ensure comparability between treated and control firms. Consequently, our baseline regression analysis findings are robust.

5. Discussion and conclusion

5.1. Principal findings, corroboration, and theoretical contributions

China's environmental policies and solutions carry substantial influence given the country's status as the world's largest emitter of greenhouse gases and its position as the largest developing economy (Wang et al., 2024a). This research's central proposition is that AI adoption is significantly associated with a reduction in CCR. Drawing on the resource-based theory (RBV), this study explores this link by examining AI's impact on firm-level climate change risk using Chinese listed firms from 2008 to 2022. The analysis confirms the hypothesis that AI adoption is robustly linked to lower CCR. This finding aligns with research emphasizing innovative technology's importance in sustainability performance (Xie et al., 2018).

AI, as evidenced by the study's results, offers several mechanisms to mitigate firm-level climate change risk. It enhances transparency and accountability enabling precise tracking and reporting through real-time data collection and analysis (Felzmann et al., 2020). This capability ensures that ecological data is accurately reported, reducing the likelihood of misreporting or manipulation. AI systems facilitate the efficient and accurate management of extensive environmental data, ensuring comprehensive monitoring practices align with stated ecological policies. Leveraging AI significantly enables firms to enhance transparency, accountability, and overall environmental performance. This technological integration provides accurate, real-time data and robust compliance mechanisms, making it harder for companies to overlook or underestimate ecological risks. Consequently, AI adoption is crucial in helping reduce firm-level climate change risk and promoting authentic corporate environmental responsibility.

Furthermore, AI's predictive capabilities are instrumental in reducing climate change risk. Predictive analytics can forecast environmental impacts and optimize resource usage, allowing firms to proactively address potential ecological issues (Nazir et al., 2024). Additionally, AI automates environmental reporting and auditing, ensuring consistency and accuracy while minimizing human error (Khan et al., 2024). Automated systems can cross-check data from multiple sources, enhancing environmental performance by providing reliable and precise information. AI-driven tools also improve stakeholder communication by providing transparent, real-time updates on sustainability efforts, fostering trust (Wamba-Taguimdje et al., 2020). Real-time monitoring of environmental parameters enables firms to generate accurate reports, reducing the temptation to falsify or embellish sustainability achievements. AI algorithms can detect anomalies and inconsistencies in ecological data, indicating potential areas of concern that require attention. Moreover, AI helps firms remain compliant with environmental regulations by continuously monitoring compliance requirements and ensuring adherence to legal standards, thus reducing the risk of non-compliance. Integrating environmental metrics into business

decision-making processes ensures that sustainability considerations are embedded into the firm's core operations. AI optimization of supply chains further reduces environmental impact, leading to genuine improvements in sustainability performance. Our findings align with the theoretical argument that AI adoption may mitigate CCR through multiple pathways. First, AI-driven data analytics can enhance firms' ability to monitor and optimize energy consumption (e.g., smart grids, predictive maintenance), reducing carbon intensity (Kaplan and Haenlein, 2021). Second, machine learning algorithms improve supply chain resilience by minimizing waste and overproduction, indirectly lowering Scope 3 emissions (Belhadi et al., 2023). Third, AI-powered climate modeling could help firms anticipate physical risks (e.g., extreme weather) and adapt proactively.

The significant reduction in firm-level climate change risk due to AI adoption in Chinese SOEs compared to non-SOEs can be attributed to factors such as closer government ties, stricter regulatory oversight, greater financial and institutional resources, and more formalized internal processes. These factors enable SOEs to effectively leverage AI technologies to improve their environmental management and mitigate climate change risks. One possible reason why adopting AI significantly reduces firm-level climate change risk in state-owned enterprises (SOEs) compared to non-SOEs in the context of China could be the distinct governance and operational structures that characterize SOEs in the country.

Firstly, SOEs in China often have closer ties with the government and greater access to state resources, which can facilitate the integration of advanced technologies like AI. This relationship allows SOEs to benefit from government initiatives and subsidies, promoting technological advancements and environmental sustainability (Lee et al., 2017). The government's strong push for digital transformation and ecological protection can provide SOEs with the necessary support and incentives to adopt AI solutions more rapidly and effectively than non-SOEs. Secondly, SOEs are typically subject to stricter regulatory oversight and accountability measures. The Chinese government imposes rigorous environmental regulations on SOEs to ensure they meet national and international environmental standards. As a result, SOEs are more likely to adopt AI technologies to comply with these regulations, enhance their environmental performance, and mitigate climate change risks. The pressure to meet regulatory requirements and maintain a positive public image can drive SOEs to leverage AI for better environmental management and reporting.

Furthermore, SOEs' larger scale and resources enable them to invest in cutting-edge AI technologies and infrastructure (Girma et al., 2009). SOEs often have the financial capacity and institutional support to implement comprehensive AI systems that monitor, analyze, and manage environmental impacts effectively. This capacity allows SOEs to deploy AI-driven solutions for optimizing resource usage, predicting ecological risks, and ensuring compliance with environmental standards, reducing climate change risks more significantly than non-SOEs.

One of our results suggests a strong relationship between AI adoption and decreased CCR in firms belonging to polluting industries, more than in non-polluting industries. This phenomenon can be attributed to several reasons. Polluting industries inherently have higher environmental impacts and, thus, a greater scope for improvement (Hussain et al., 2022). Adopting AI in these industries can significantly reduce emissions and waste, as AI technologies optimize processes, reduce inefficiencies, and predict potential environmental risks before they escalate. Polluting industries are often subject to stricter environmental regulations and scrutiny (Chang et al., 2015). Firms in these industries may adopt AI technologies to ensure compliance with environmental standards, avoid penalties, and improve their public image. AI can provide precise monitoring and reporting capabilities, helping firms stay within regulatory limits and demonstrate their commitment to sustainability.

Government initiatives and support play a crucial role in China's specific context. The Chinese government has been aggressively

promoting environmental sustainability and the adoption of AI across industries.

Polluting industries in China may receive more government support and incentives to adopt AI technologies as part of national efforts to reduce pollution and combat climate change. This support can include subsidies, tax breaks, and favorable policies, making it easier for polluting firms to implement AI solutions. Furthermore, China has implemented stringent environmental regulations, particularly targeting heavily polluting industries. These regulations compel firms to adopt advanced technologies like AI to comply with ecological standards and avoid severe penalties. The regulatory environment in China thus drives the adoption of AI in polluting industries to ensure compliance and enhance environmental performance. There is also increasing public awareness and concern about environmental issues in China. Polluting industries face significant pressure from the public and the market to improve their environmental practices. Adopting AI can help these firms meet societal expectations, improve their reputation, and gain a competitive advantage by demonstrating their commitment to sustainability. Lastly, the Chinese government is focused on upgrading its industrial sector by integrating advanced technologies, including AI. Polluting industries are a key target for this upgradation due to their significant environmental impact. AI adoption is encouraged as part of broader efforts to modernize these industries, improve their efficiency, and reduce their ecological footprint. These factors collectively drive the adoption of AI in polluting industries, leading to substantial improvements in environmental performance and reductions in climate change risk.

The significant negative correlation between firm-level climate change risk and AI adoption in high-tech industries can be attributed to several key factors. High-tech firms, with their inherent focus on innovation and advanced technological capabilities, are in a prime position to leverage AI to address environmental challenges. These companies have the necessary technological infrastructure and the organizational ability to implement sophisticated AI systems. For instance, AI can optimize resource management, develop eco-friendly products, and enhance energy efficiency, which are critical in managing climate-related risks effectively.

The high-tech industries' technological environment facilitates AI adoption by providing the necessary tools and expertise for its application. These sectors are known for their rapid technological advancements and are strongly inclined to integrate cutting-edge solutions. This aligns with the TOE paradigm, which posits that organizations in technologically advanced contexts are better positioned to adopt and implement innovative technologies like AI for managing climate change risks (Tornatzky et al., 1990). Moreover, the pressure from stakeholders—such as investors, consumers, and regulatory bodies—compels high-tech firms to adopt environmentally responsible practices. This external pressure enhances the motivation for these firms to deploy AI technologies to meet environmental standards and mitigate climate change impacts. The availability of resources and a knowledge base in high-tech industries further supports the practical application of AI in analyzing data, conducting predictive analyses, and guiding risk management decisions. Finally, sector-specific evaluations highlight AI has transformative potential for reducing emissions, particularly in innovative fields like manufacturing and information technology (Cowls et al., 2023; Dauvergne, 2022; Pham et al., 2024). This underscores the TOE framework's assertion that high-tech industries' organizational, technological, and environmental contexts drive the effective integration of AI for climate risk management.

AI adoption is significantly associated with a reduction in CCR in companies with high board independence, which can be attributed to several factors rooted in governance and oversight dynamics. Board independence is crucial in shaping the strategic decisions related to AI implementation, especially for managing climate change risks. Independent boards are better equipped to oversee and monitor management's decisions without the influence of internal biases or conflicts of

interest. This autonomy allows them to ensure that AI technologies are adopted for short-term gains and long-term sustainability and risk management. The agency theory provides a valuable framework for understanding this dynamic. It posits that conflicts of interest between shareholders and management can lead to agency costs when managers prioritize their objectives over those of shareholders (Kyere and Ausloos, 2021). By their detachment from day-to-day operations, independent directors are more likely to align managerial actions with shareholder interests, particularly in areas critical to sustainability like climate change risk management. High board independence enhances the scrutiny and effectiveness of strategic decisions related to AI adoption. Independent directors must enforce rigorous oversight, ensuring that AI technologies are effectively chosen and implemented to mitigate climate-related risks. Their role in evaluating and endorsing technology investments aligns with the shareholders' goals of reducing future risks and promoting corporate sustainability (Hilb, 2020; Todaro et al., 2021). This oversight helps firms leverage AI to enhance their risk detection, assessment, and management capabilities, thus lowering their exposure to climate change risks. Moreover, empirical research supports that board independence positively influences corporate sustainability practices, including AI adoption for climate risk management. Independent boards are perceived as more responsible and diligent in representing the interests of various stakeholders, including shareholders, employees, customers, and the public. This broader perspective enables them to support AI initiatives that reduce carbon footprints and enhance environmental management (Haque, 2017; Alipour et al., 2019). As a result, firms with higher board independence are more likely to implement AI solutions effectively, leading to a more substantial reduction in climate change risks.

The robust linkage of AI adoption and climate change risk in firms with highly gender-diverse boards can be attributed to several key factors related to the benefits of diverse perspectives and improved decision-making processes. Gender diversity on boards brings together a variety of viewpoints and approaches, which can enhance the effectiveness of AI technologies in managing environmental risks. Gender diversity in corporate governance improves the firm's ability to deploy AI technologies effectively for climate risk management. By fostering a variety of viewpoints and ensuring rigorous oversight, gender-diverse boards are better positioned to implement AI solutions that mitigate climate change risks and promote sustainable practices (Eroglu and Karatepe Kaya, 2022; Zhao and Gómez Fariñas, 2023). Diverse teams, particularly those with a balanced representation of genders, are known to excel in creativity, innovation, and problem-solving. This aligns with diversity theory, suggesting that heterogeneous teams generate new ideas and tackle complex issues more effectively than homogenous groups (Díaz-Fernández et al., 2020; Wu et al., 2022). In AI adoption for climate risk management, having a gender-diverse board means the firm benefits from a broader range of perspectives and strategies. This can lead to more comprehensive and innovative approaches to reducing ecological footprints and implementing effective AI-driven solutions.

Additionally, gender diversity on boards often correlates with better oversight and scrutiny of managerial decisions related to risk management and sustainability. Diverse boards are more likely to critically evaluate and monitor AI adoption strategies, ensuring they align with the firm's climate goals and sustainability objectives (Al-Qahtani and Elgharbawy, 2020; Lin and Wu, 2023). This improved governance can enhance the effectiveness of AI technologies in addressing climate change risks by promoting thorough review processes and accountability. Furthermore, research has shown that gender-diverse management teams contribute positively to financial performance metrics, indicating that diverse perspectives can drive better outcomes across various business operations (Chijoke-Mgbame et al., 2020; Martinez-Jimenez et al., 2020). In the realm of AI and sustainability, this means that gender-diverse boards are likely to apply AI technologies in ways that are not only innovative but also equitable, taking into account a broader range of environmental issues and potential biases in algorithms

(Hickman and Petrin, 2021; Nuseir et al., 2021; Toniolo et al., 2020).

This study contributes significantly to theoretical literature in several ways. Firstly, it adopts a multi-theoretical approach to construct a framework elucidating the relationship between AI and CCL. Drawing from resource-based view theory, stakeholder theory, agency theory, and legitimacy theory, this research aims to delineate the intricate dynamics between AI and CCL theoretically. Unlike previous studies that have mainly focused on the profit impact of AI on overall firm performance, innovation, and productivity, respectively (Bahoo et al., 2023; Bosse et al., 2023; Czarnitzki et al., 2023), our study takes a different approach. We delve into the social perspective and investigate how AI impacts CCL at the firm level. By exploring this specific dimension, we establish a crucial link between AI adoption and CCL, revealing the social, economic, and environmental implications of integrating AI into businesses.

Secondly, the research establishes an empirical link between AI and CCL, resolving the ambiguity surrounding their positive or negative relationship. This study presents original empirical data substantiating the inverse correlation between artificial intelligence (AI) and CCL. This fills a significant knowledge void regarding the capacity of sophisticated digital technologies to propel authentic, environmentally sustainable performance. This discovery contributes to the knowledge on adopting artificial intelligence by emphasizing its capacity to counteract climate change. By doing so, it seeks to understand the factors influencing the impact of AI on firm-level climate change risk with varying fundamental circumstances, particularly exploring how different ownership structures may moderate this effect. Investigating whether this impact remains consistent across diverse ownership types, including comparisons between SOEs and non-SOEs and local SOEs and central SOEs, sheds light on the importance of corporate internal governance and resource allocation techniques for fostering sustainable and ecologically responsible company operations. The research uncovers significant intricacies by demonstrating that the influence of AI on CCL is more pronounced in privately owned enterprises in China than in state-owned enterprises. This observation illuminates the impact of ownership structure on reactions to novel technologies, a domain that has received limited attention in technology adoption studies.

These findings enrich the literature on innovation in emerging economies such as China, where state ownership remains prevalent. Moreover, this research offers significant contributions for policymakers, managers, and stakeholders to provide informed strategies and interventions regarding the advantages of implementing AI in sustainable business practices. Organizations can foster a more sustainable future by incorporating these insights into their policymaking, investment decisions, and stakeholder engagements, encouraging environmentally favorable digital economy practices.

From a pragmatic perspective, this study provides valuable information and suggestions for firms, decision-makers, and industry participants who want to leverage AI technologies' revolutionary capacity to promote sustainable development and climate change resilience. By clarifying industry-specific AI applications and their impact on sustainability, this research offers valuable advice to firms trying to manage the challenges of technological advancement in a rapidly evolving business climate. Additionally, to optimize the advantages of AI adoption for climate change risk management, this research emphasizes the significance of strong governance frameworks, stakeholder participation strategies, and initiatives to promote diversity. Firms may increase their ability to adapt and resilience to crises and disruptions connected to climate change by cultivating a philosophy of innovative thinking, ethical technological use, and inclusive decision-making procedures.

While AI aids in climate risk mitigation, its environmental costs—particularly high energy use and e-waste—must be addressed. Training large AI models can emit hundreds of tonnes of CO₂, while rapid hardware turnover exacerbates electronic waste. Sustainable solutions, including renewable-powered data centers (e.g., Google's 24/7 carbon-free energy pledge), next-generation nuclear energy like small

modular reactors, and energy-efficient hardware designs, are emerging. Balancing AI's climate benefits with its footprint requires prioritizing clean energy, efficient computing, and responsible hardware lifecycle management.

5.2. Conclusion, implications, limitations, and future recommendations

5.2.1. Conclusion

Our findings indicate that adopting artificial intelligence (AI) reduces firm-level climate change risk. This effect is notably more pronounced in state-owned enterprises, firms with gender-diverse and independent boards, and those operating within high-tech and polluting industries. Conversely, the impact of AI adoption on mitigating climate change risk is less significant in non-state-owned firms, those with less gender diversity and non-independent boards, and companies situated in non-polluting and non-high-tech sectors.

This study demonstrates that artificial intelligence adoption is an effective tool for controlling firm-level climate change risk, with powerful effects observed in state-owned enterprises, pollution-intensive industries, and firms with robust corporate governance structures. Our findings reveal that AI's climate risk-reduction potential is not uniform but context-dependent, varying significantly across ownership types, sectors, and governance characteristics. The results highlight how technological solutions interact with organizational factors to shape environmental outcomes, suggesting that AI adoption must be strategically aligned with a firm's specific institutional and operational context. By establishing these patterns through rigorous empirical analysis, we provide a methodological framework and substantive findings that advance understanding of digital technologies' role in corporate climate resilience. The evidence underscores AI's growing importance as part of the business sector's comprehensive climate risk management strategies.

5.2.2. Implications

The findings from this research offer significant implications for managers, shareholders, government entities, and other stakeholders concerning adopting artificial intelligence (AI) for climate change risk management. For managers, the results highlight the importance of fostering organizational characteristics that enhance the effectiveness of AI in mitigating climate change risks. Managers in state-owned enterprises and those with gender-diverse, independent boards are better positioned to leverage AI for environmental sustainability. Consequently, enhancing board diversity and independence could be a strategic focus for improving the efficacy of AI initiatives. Furthermore, managers in high-tech and polluting industries should prioritize AI adoption as a crucial tool for better climate risk management, given its pronounced impact in these contexts. Shareholders can benefit from understanding how AI adoption interacts with firm characteristics to influence climate risk management. Investing in state-owned companies that have gender-diverse and independent boards, and operate in high-tech or polluting industries, may offer greater assurance of effective climate risk mitigation through AI. This insight can guide investment decisions and advocacy for improved governance practices within portfolio companies, ensuring that investments are aligned with sustainability goals. Government policymakers and regulatory bodies can utilize these findings to craft targeted interventions that encourage AI adoption for climate change mitigation. Policies that promote board diversity and independence, as well as support for high-tech and polluting industries in integrating advanced technologies, could amplify the benefits of AI in addressing climate risks.

Additionally, governments might consider providing incentives or support for state-owned enterprises to implement AI solutions for sustainability, further enhancing their climate risk management capabilities. For other stakeholders, including consumers, environmental organizations, and industry groups, these findings underscore the importance of advocating for and supporting practices that maximize

the impact of AI on climate change risk management. Encouraging firms to adopt AI technologies, especially in sectors with more pronounced effects, and supporting governance reforms that foster greater board diversity and independence can help drive industry-wide adoption of AI for environmental sustainability.

5.2.3. Limitations and future recommendations

While this study provides valuable insights into the relationship between AI adoption and FLCCR in Chinese listed firms, it is important to acknowledge several inherent limitations. First, the fact that our sample consists only of Chinese listed companies may introduce bias compared to unlisted companies with different traits, outcomes, or behaviors. Given that our sample only includes listed companies, it may not fairly reflect all Chinese industries or sectors. As a result, our results might not provide a thorough insight or consider other company types. A more comprehensive analysis might be possible by including unlisted companies from broader industries.

Second, our findings are based exclusively on firms operating in China, which may limit their direct generalizability to other national or regional contexts. China's unique cultural norms (e.g., collectivist business practices), institutional structures (e.g., state-led economic policies), and regulatory environment (e.g., centralized environmental governance) could significantly influence the AI-FLCCR relationship in ways that may not apply to firms in liberal-market economies or other emerging markets. For instance, the Chinese government's active role in promoting AI development and carbon neutrality may create less pronounced synergies in countries with weaker state intervention. Similarly, the digital market in China is characterized by distinct platform ecosystems, data governance policies, and innovation incentives that differ from those in the West or other Asian economies. These contextual factors suggest that our findings should be interpreted cautiously when extrapolating to other settings. Future research should systematically examine the AI-CCR relationship in a variety of countries and institutional settings (e.g., developed economies, other emerging economies, economies with different governance models) to rigorously assess the cross-cultural and cross-institutional generalisability (or lack thereof) of the current findings and to identify potential contextual moderators that were not considered in this study.

Our findings demonstrate a robust association between AI adoption and reduced climate change risk (CCR), but this relationship should be interpreted cautiously. While plausible mechanisms exist, our empirical design cannot establish direct causation. The observed link may reflect broader digital transformation efforts or unobserved firm capabilities. Although we address endogeneity through fixed effects models and robustness checks, reverse causality remains a concern—firms with higher CCR may invest more in AI for risk mitigation. Additionally, omitted variables (e.g., concurrent ESG initiatives) could influence AI adoption and CCR reductions.

While textual analysis of MD&A sections offers a valuable approach to studying AI adoption, its limitations are noteworthy. Firms often strategically disclose AI-related information, emphasizing socially desirable narratives while omitting less favorable details. This selective disclosure means textual data cannot accurately capture the actual breadth, depth, or implementation intensity of AI adoption, whether tools are merely mentioned, in pilot phases, or fully integrated. Our computer-aided text analysis of MD&As shares these constraints, as companies may present curated narratives that limit insights into real AI usage. Moreover, while methodologically practical for large-scale analysis, our binary AI adoption measure presents an essential limitation by oversimplifying implementation realities. This approach cannot capture variations in AI sophistication (basic automation vs. advanced ML), deployment scale (pilots vs. enterprise-wide), or functional breadth (single vs. multi-department use), potentially obscuring nuanced relationships with climate risk management. These constraints should be considered when interpreting our findings. Future research should employ complementary methods, such as longitudinal case studies of AI-

implementing firms or integration of non-textual proxies (e.g., AI investments, employee skills), to provide a more comprehensive understanding of AI adoption patterns.

Our study acknowledges the inherent complexity of organizational contexts in shaping the AI-carbon reduction relationship. While we have examined several essential moderators, including ownership structure, industry characteristics, and governance quality, we recognize that other influential organizational factors remain unexplored in our current analysis. These may include the alignment between organizational culture and sustainability objectives, managerial attitudes and cognitive frameworks regarding environmental risks and technological solutions, the specific types of AI applications implemented (such as machine learning for predictive analytics versus computer vision for environmental monitoring), as well as variations in organizational capabilities for adopting and deploying AI technologies. These unexamined dimensions likely play significant roles in mediating the effectiveness of AI for climate risk mitigation, suggesting our findings represent a critical but partial understanding of this complex phenomenon.

Building on these insights, future research would benefit from systematically investigating this broader range of potential moderating variables. Particularly valuable would be studies that explore how different organizational cultures influence AI implementation for sustainability goals, examine the role of leadership perspectives in driving technological solutions for environmental challenges, analyze how specific AI applications yield varying carbon reduction outcomes, and assess how organizational learning capacity affects AI adoption success. Such multidimensional inquiries, combining strategic management perspectives with insights from cognitive psychology and technology studies, could significantly advance our understanding of the organizational conditions under which AI most effectively contributes to climate change mitigation. This would address current limitations and provide more nuanced guidance for practitioners seeking to leverage AI for sustainability objectives. Additionally, media sources, foreign auditors, Big Four auditors, and other monitoring organizations may impact the relationship between AI and climate change risk. More organizational-level moderators should be investigated in future studies to offer a more comprehensive picture.

Despite our use of advanced econometric techniques and rigorous robustness checks, this observational study faces inherent limitations in establishing definitive causal relationships. While our baseline analyses provide evidence supporting the association between AI adoption and carbon reduction, we cannot completely rule out potential endogeneity concerns. These include the possibility of reverse causality, where firms already committed to sustainability may be more likely to adopt AI, or unobserved confounding factors such as unmeasured management practices or external market forces that could influence both AI implementation and carbon performance simultaneously. Although we have taken steps to address these issues through Two-stage Heckman tests, Generalized Method of Moments tests, Univariate Analysis, Propensity Score Matching (PSM), and Entropy Balancing tests, these methods cannot fully replicate the controlled conditions of experimental research. Looking ahead, future research could build on our findings by

pursuing designs that move closer to causal identification. One promising direction would be to leverage natural experiments, such as sudden policy changes or regional variations in AI infrastructure, that create exogenous shocks to AI adoption patterns. Another approach could involve in-depth longitudinal case studies tracking firms before and after AI implementation to establish temporal ordering better and isolate AI's specific effects. Field experiments conducted with corporations could also provide valuable insights by randomly assigning AI tools across different operational units while controlling for confounding variables. Additionally, future work might explore more sophisticated instrumental variables that satisfy stricter exclusion criteria, such as regulatory thresholds or geographic factors that influence AI uptake but are unlikely to affect carbon outcomes directly. These methodological advances would help strengthen confidence in the causal mechanisms underlying the AI-carbon reduction relationship while building upon the foundational evidence provided by our study.

While this study provides valuable insights into the relationship between AI adoption and climate risk management in Chinese firms from 2008 to 2022, it primarily captures the early stages of AI integration. Given that AI technologies were still maturing during this period, their full and potentially transformative impact on corporate climate resilience (CCR) may not be fully realized. The observed effects may represent initial adaptations rather than long-term systemic changes, as deeper AI integration into core business processes and broader industry adoption could unfold over longer time horizons. Future research could extend this work by adopting a longer-term perspective, tracking firms over multiple decades to assess how sustained AI adoption influences climate risk management. Longitudinal studies would help capture the evolving role of AI as it becomes more sophisticated, widely diffused across industries, and embedded into strategic decision-making.

Additionally, such research could explore whether early AI adopters gain a competitive advantage in climate resilience compared to late adopters, and how regulatory and technological advancements shape these dynamics over time. Furthermore, while corporate disclosures provide valuable breadth of coverage, they may be subject to reporting biases wherein firms overstate AI adoption or underrepresent climate risks. An auspicious approach would combine self-reported disclosures with operational indicators like data center energy consumption or third-party verified performance metrics to enhance reliability.

CRediT authorship contribution statement

Muhammad Kaleem Khan: Conceptualization, Writing – original draft. **Muhammad Jameel Hussain:** Formal analysis, Data curation. **Muhammad Wasim Hussan:** Writing - original draft. **Anona Armstrong:** Conceptualization. **Shanshan Li:** Resources, Data curation, Supervision. **Afifa Qadeer:** Writing – original draft.

Declaration of competing interest

The authors declare no conflict of interest.

Appendix

Variables used in the model and their definitions.

AI	Artificial Intelligence adoption level obtained through textual analysis
FLCCR	Firm Level Climate Change Risk obtained through textual analysis
Board Size	Total numbers of board of directors
Board_Ind	Total numbers of independent directors divided by total numbers of directors
CEO Duality	Dummy variable equal to 1, if CEO also holds the chairman position
CEO Gender	Dummy variable equal to 1, if CEO is male and zero otherwise.

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ROA	Net profit divided by total assets
Firm Size	Taking log of total assets
Firm Age	Number of years since firm is listed on stock exchange
Leverage	Total debt divided by total assets
Top10	Percentage of ownership of top ten shareholders
Cap_Intensity	Ratio of capital expenditures divided by total assets.
R&D Intensity	Ratio of R&D expenditure divided by total assets.
SOE	Dummy variable equals to 1, if firms ownership belongs to state and zero otherwise.
Polluting firms	Dummy variable equals to 1, if firms belong to polluting industries sectors and zero otherwise.
High-tech	Dummy variable equals to 1, if firms belong to high-tech industries sectors and zero otherwise.
FLCCR	Firm-Level Climate Change Risk

Data availability

Data will be made available on request.

References

- Abbasi, S., Erdebili, B., 2023. Green closed-loop supply chain networks' response to various carbon policies during COVID-19. *Sustainability* 15 (4), 3677.
- Abhayawansa, S., Adams, C.A., Neesham, C., 2021. Accountability and governance in pursuit of sustainable development goals: conceptualising how governments create value. *Account. Audit. Account. J.* 34 (4), 923–945. <https://doi.org/10.1108/AAAJ-07-2020-4667>.
- Aghion, P., 2018. Innovation and growth from a Schumpeterian perspective. *Rev. Econ. Polit.* 128 (5), 693–711.
- Agoraki, K.K., Giaka, M., Konstantios, D., Negkakos, I., 2024. The relationship between firm-level climate change exposure, financial integration, cost of capital and investment efficiency. *J. Int. Money Financ.* 141, 102994.
- Ahmad, T., Zhu, H., Zhang, D., Tariq, R., Bassam, A., Ullah, F., AlGhamdi, A.S., Alshamrani, S.S., 2022. Energetics systems and artificial intelligence: applications of industry 4.0. *Energy Rep.* 8, 334–361.
- Ai, L., Gao, L.S., 2023. Firm-level risk of climate change: evidence from climate disasters. *Glob. Financ. J.* 55, 100805.
- Akomea-Frimpong, I., Dzagli, J.R.A.D., Eluerkeh, K., Bonsu, F.B., Opoku-Brafi, S., Gyimah, S., Kukah, A.S., 2023. A systematic review of artificial intelligence in managing climate risks of PPP infrastructure projects. *Eng. Constr. Archit. Manag.* (ahead-of-print).
- Albitar, K., Al-Shaer, H., Liu, Y.S., 2023. Corporate commitment to climate change: the effect of eco-innovation and climate governance. *Res. Policy* 52 (2), 104697. <https://doi.org/10.1016/j.respol.2022.104697>.
- Alekseeva, L., Gine, M., Samila, S., Taska, B., 2020. AI Adoption and Firm Performance: Management Versus IT. Available at SSRN 3677237.
- Alipour, M., Ghanbari, M., Jamshidinavid, B., Taherabadi, A., 2019. Does board independence moderate the relationship between environmental disclosure quality and performance? Evidence from static and dynamic panel data. *Corp. Gov. Int. J. Bus. Soc.* 19 (3), 580–610. <https://doi.org/10.1108/CG-06-2018-0196>.
- Al-Qahtani, M., Elgharabawy, A., 2020. The effect of board diversity on disclosure and management of greenhouse gas information: evidence from the United Kingdom. *J. Enterp. Inf. Manag.* 33 (6), 1557–1579. <https://doi.org/10.1108/JEIM-08-2019-0247>.
- Ansari, S.P.M.A., Odongo, J.O., Nomani, M.Z.M., Salahuddin, G., Hasan, M.F., Pallathadka, L.K., 2022. Evaluating the role of environment management system based on artificial intelligence. *Mater. Today Proc.* 56, 2240–2244. <https://doi.org/10.1016/j.matpr.2021.11.571>.
- Babina, T., Fedyk, A., He, A.X., Hodson, J., 2020. Artificial Intelligence, Firm Growth, and Industry Concentration. SSRN Scholarly Paper ID 3651052.. Social Science Research Network, Rochester, NY.
- Bahoo, S., Cucculelli, M., Qamar, D., 2023. Artificial intelligence and corporate innovation: a review and research agenda. *Technol. Forecast. Soc. Chang.* 188, 122264.
- Barton, D., Woetzel, J., Seong, J., Tian, Q., 2017. Artificial intelligence: implications for China. <http://dl.n.jaipuria.ac.in:8080/jspui/bitstream/123456789/1888/1/MGI-Artificial-intelligence-implications-for-China.pdf>.
- Belhadi, A., Kamble, S., Jabbour, C.J.C., Gunasekaran, A., Ndubisi, N.O., Venkatesh, M., 2023. Manufacturing and service supply chain resilience to the COVID-19 outbreak: lessons learned from the automobile and airline industries. *Technol. Forecast. Soc. Chang.* 163, 120447. <https://doi.org/10.1016/j.techfore.2020.120447>.
- Bell, E., Harley, B., Bryman, A., 2022. *Business Research Methods*. Oxford university press.
- Bendig, D., Schulz, C., Theis, L., Raff, S., 2023. Digital orientation and environmental performance in times of technological change. *Technol. Forecast. Soc. Chang.* 188, 122272.
- Benzidia, S., Makaoui, N., Bentahar, O., 2021. The impact of big data analytics and artificial intelligence on green supply chain process integration and hospital environmental performance. *Technol. Forecast. Soc. Chang.* 165, 120557.
- Boero, R., 2024. An AI-enhanced systematic review of climate adaptation costs: approaches and advancements, 2010–2021. *Climate* 12 (8), 116.
- Bolton, P., Kacperczyk, M., 2021. Do investors care about carbon risk? *J. Financ. Econ.* 142 (2), 517–549.
- Bosse, D., Thompson, S., Ekman, P., 2023. In consilium apparatus: artificial intelligence, stakeholder reciprocity, and firm performance. *J. Bus. Res.* 155, 113402.
- Bu, C., Shi, D., 2021. The emission reduction effect of daily penalty policy on firms. *J. Environ. Manag.* 294, 112922. <https://doi.org/10.1016/j.jenvman.2021.112922>.
- Burrell, G., Morgan, G., 2019. *Sociological Paradigms and Organisational Analysis: Elements of the Sociology of Corporate Life*. Routledge.
- Burke, J.J., 2022. Do boards take environmental, social, and governance issues seriously? Evidence from media coverage and CEO dismissals. *J. Bus. Ethics* 176 (4), 647–671. <https://doi.org/10.1007/s10551-020-04715-x>.
- Chang, L., Li, W., Lu, X., 2015. Government engagement, environmental policy, and environmental performance: evidence from the most polluting Chinese listed firms. *Bus. Strateg. Environ.* 24 (1), 1–19.
- Chava, S., 2014. Environmental externalities and cost of capital. *Manag. Sci.* 60 (9), 2223–2247.
- Cheng, J., Zeng, J., 2023. Shaping AI's future? China in global AI governance. *J. Contemp. China* 32 (143), 794–810.
- Chijoke-Mgbame, A.M., Boateng, A., Mgbame, C.O., 2020. Board gender diversity, audit committee and financial performance: evidence from Nigeria. *Account. Forum* 44 (3), 262–286. <https://doi.org/10.1080/01559982.2020.1766280>.
- Cowls, J., Tsamados, A., Taddeo, M., Floridi, L., 2023. The AI gambit: leveraging artificial intelligence to combat climate change—opportunities, challenges, and recommendations. *AI Soc.* 38 (1), 283–307. <https://doi.org/10.1007/s00146-021-01294-x>.
- Czarnitzki, D., Fernández, G.P., Rammer, C., 2023. Artificial intelligence and firm-level productivity. *J. Econ. Behav. Organ.* 211, 188–205.
- Dannouni, A., Deutscher, S.A., Dezzaz, G., Elman, A., Gawel, A., Hanna, M., Hyland, A., Kharij, A., Maher, H., Patterson, D., Jones, E.R., Rothenberg, J., Tber, H., Texier, M., Ziat, A., 2023. Accelerating Climate Action with AI. Technical Report. Boston Consulting Group Commissioned by Google. <https://web-assets.bcg.com/72/cf/b609ac3d4ac6829bae6fa88b8329/bcg-accelerating-climate-action-with-ainov-2023-rev.pdf>.
- Dauvergne, P., 2022. Is artificial intelligence greening global supply chains? Exposing the political economy of environmental costs. *Rev. Int. Polit. Econ.* 29 (3), 696–718.
- de Grosbois, D., Fennell, D.A., 2022. Determinants of climate change disclosure practices of global hotel companies: application of institutional and stakeholder theories. *Tour. Manag.* 88, 104404. <https://doi.org/10.1016/j.tourman.2021.104404>.
- Delmas, M., Toffel, M.W., 2004. Stakeholders and environmental management practices: an institutional framework. *Bus. Strateg. Environ.* 13 (4), 209–222.
- Dessaint, O., Matray, A., 2017. Do managers overreact to salient risks? Evidence from hurricane strikes. *J. Financ. Econ.* 126 (1), 97–121.
- Díaz-Fernández, M.C., González-Rodríguez, M.R., Simonetti, B., 2020. Top management team diversity and high performance: an integrative approach based on upper echelons and complexity theory. *Eur. Manag. J.* 38 (1), 157–168. <https://doi.org/10.1016/j.emj.2019.06.006>.
- Dinga, C.D., Wen, Z., 2022. China's green deal: can China's cement industry achieve carbon neutral emissions by 2060? *Renew. Sust. Energ. Rev.* 155, 111931.
- Elsayih, J., Datt, R., Tang, Q., 2021. Corporate governance and carbon emissions performance: empirical evidence from Australia. *Aust. J. Environ. Manag.* 28 (4), 433–459. <https://doi.org/10.1080/14486563.2021.1989066>.
- Eroglu, M., Karatepe Kaya, M., 2022. Impact of artificial intelligence on corporate board diversity policies and regulations. *Eur. Bus. Organ. Law Rev.* 23 (3), 541–572. <https://doi.org/10.1007/s40804-022-00251-5>.
- Felzmann, H., Fosch-Villaronga, E., Lutz, C., Tamó-Larriex, A., 2020. Towards transparency by design for artificial intelligence. *Sci. Eng. Ethics* 26 (6), 3333–3361.
- Feng, C., Ye, X., Li, J., Yang, J., 2024. How does artificial intelligence affect the transformation of China's green economic growth? An analysis from internal-structure perspective. *J. Environ. Manag.* 351, 119923.
- Ferdous, L.T., Atawnah, N., Yeboah, R., Zhou, Y., 2024. Firm-level climate risk and accounting conservatism: international evidence. *Int. Rev. Financ. Anal.* 95, 103511.
- Galaz, V., Centeno, M.A., Callahan, P.W., Causevic, A., Patterson, T., Brass, I., Baum, S., Farber, D., Fischer, J., Garcia, D., McPhearson, T., Jimenez, D., King, B., Larcey, P., Levy, K., 2021. Artificial intelligence, systemic risks, and sustainability. *Technol. Soc.* 67, 101741. <https://doi.org/10.1016/j.techsoc.2021.101741>.
- Ghafoor, A., Sêho, M., Sifat, I., 2023. Co-opted board and firm climate change risk. *Financ. Res. Lett.* 52, 103508. <https://doi.org/10.1016/j.frl.2022.103508>.

- Giglio, S., Kelly, B., Stroebe, J., 2021. Climate finance. *Annu. Rev. Financ. Econ.* 13 (1), 15–36.
- Girma, S., Gong, Y., Görg, H., 2009. What determines innovation activity in Chinese state-owned enterprises? The role of foreign direct investment. *World Dev.* 37 (4), 866–873.
- Goodell, J.W., Palma, A., Paltrinieri, A., Piserà, S., 2025. Firm-level climate change risk and corporate debt maturity. *J. Int. Money Financ.* 152, 103275.
- Goralski, M.A., Tan, T.K., 2020. Artificial intelligence and sustainable development. *Int. J. Manag. Educ.* 18 (1), 100330.
- Han, H., Shiwakoti, R.K., Jarvis, R., Mordi, C., Botchie, D., 2023. Accounting and auditing with blockchain technology and artificial intelligence: a literature review. *Int. J. Account. Inf. Syst.* 48, 100598. <https://doi.org/10.1016/j.accinf.2022.100598>.
- Haque, F., 2017. The effects of board characteristics and sustainable compensation policy on carbon performance of UK firms. *Br. Account. Rev.* 49 (3), 347–364. <https://doi.org/10.1016/j.bar.2017.01.001>.
- Hickman, E., Petrin, M., 2021. Trustworthy AI and corporate governance: the EU's ethics guidelines for trustworthy artificial intelligence from a company law perspective. *Eur. Bus. Organ. Law Rev.* 22 (4), 593–625. <https://doi.org/10.1007/s40804-021-00224-0>.
- Hilb, M., 2020. Toward artificial governance? The role of artificial intelligence in shaping the future of corporate governance. *J. Manag. Gov.* 24 (4), 851–870. <https://doi.org/10.1007/s10997-020-09519-9>.
- Hoang, K., Pham, L., Ha, O.K., Nghiem, H.X., 2025. Firm-level climate change risk exposure and firm efficiency. *Int. Rev. Econ. Financ.* 101, 104220. <https://doi.org/10.1016/j.iref.2025.104220>.
- Hollander, S., Pronk, M., Roelofs, E., 2010. Does silence speak? An empirical analysis of disclosure choices during conference calls. *J. Account. Res.* 48 (3), 531–563.
- Hong, H., Li, F.W., Xu, J., 2019. Climate risks and market efficiency. *J. Econ.* 208 (1), 265–281.
- Hossain, A., Masum, A.A., Saadi, S., Benkraiem, R., Das, N., 2023. Firm-level climate change risk and CEO equity incentives. *Br. J. Manag.* 34 (3), 1387–1419.
- Hossain, A.T., Attig, N., Hebb, G., Hasan, M., 2025. Firm-level litigation risk and CEO equity incentives. *Financ. Res. Lett.* 78, 107151.
- Hussain, M.J., Tian, G., Ayaz, M., Zhang, X., 2022. The impact of directors' foreign experience on environmental information disclosure: evidence from heavily polluting Chinese firms. *Asia Pac. J. Financ. Stud.* 51 (3), 486–509.
- Huynh, T.D., Xia, Y., 2021. Climate change news risk and corporate bond returns. *J. Financ. Quant. Anal.* 56 (6), 1985–2009.
- Jain, H., Dhupher, R., Shrivastava, A., Kumar, D., Kumari, M., 2023. AI-enabled strategies for climate change adaptation: protecting communities, infrastructure, and businesses from the impacts of climate change. *Comput. Urban Sci.* 3 (1), 25. <https://doi.org/10.1007/s43762-023-00100-2>.
- Jamali, D., 2008. A stakeholder approach to corporate social responsibility: a fresh perspective into theory and practice. *J. Bus. Ethics* 82, 213–231.
- Javadi, S., Masum, A.-A., 2021. The impact of climate change on the cost of bank loans. *Finance* 69, 102019.
- Jensen, M.C., Meckling, W.H., 1976. Theory of the firm: managerial behavior, agency costs and ownership structure. *J. Financ. Econ.* 3 (4), 305–360. [https://doi.org/10.1016/0304-405X\(76\)90026-X](https://doi.org/10.1016/0304-405X(76)90026-X).
- Jia, Z., Lin, B., 2021. How to achieve the first step of the carbon-neutrality 2060 target in China: the coal substitution perspective. *Energy* 233, 121179.
- Jinping, X., 2022. Hold High the Great Banner of Socialism with Chinese Characteristics and Strive in Unity to Build a Modern Socialist Country in all Respects. Report at the 20th National Congress of the Communist Party of China. Qiu Shi.
- Jones, A., Kuehnert, J., Fraccaro, P., Meuriot, O., Ishikawa, T., Edwards, B., Assefa, S., 2023. AI for climate impacts: applications in flood risk. *NPJ Clim. Atmos. Sci.* 6 (1), 63.
- Kaplan, A., Haenlein, M., 2021. Rulers of the world, unite! The challenges and opportunities of artificial intelligence. *Bus. Horiz.* 63 (1), 37–50.
- Katal, A., Dahiya, S., Choudhury, T., 2023. Energy efficiency in cloud computing data centers: a survey on software technologies. *Clust. Comput.* 26 (3), 1845–1875. <https://doi.org/10.1007/s10586-022-03713-0>.
- Kelly, D.L., Kolstad, C.D., 1999. Bayesian learning, growth, and pollution. *J. Econ. Dyn. Control.* 23 (4), 491–518.
- Khan, M.K., Huo, C., Zahid, R.A., Maqsood, U.S., 2024. The automated sustainability auditor: does artificial intelligence curtail greenwashing behavior in Chinese firms? *Bus. Strateg. Environ.* 33 (8), 9015–9039.
- Khan, M. K., Hussain, P., Hussain, M. J., Michael, K. (2025). Innovation at the nexus of technology and sustainability: CEO leadership and digital transformation in China. *Sustain. Futures* 9 (2025) 100494.
- Khanal, S., Zhang, H., Taeiagh, A., 2025. Development of new generation of artificial intelligence in China: when Beijing's global ambitions meet local realities. *J. Contemp. China* 34 (151), 19–42.
- Kurniawan, T.A., Othman, M.H.D., Liang, X., Goh, H.H., Gikas, P., Kusworo, T.D., Anouzla, A., Chew, K.W., 2023. Decarbonization in waste recycling industry using digitalization to promote net-zero emissions and its implications on sustainability. *J. Environ. Manag.* 338, 117765. <https://doi.org/10.1016/j.jenvman.2023.117765>.
- Kyere, M., Ausloos, M., 2021. Corporate governance and firms financial performance in the United Kingdom. *Int. J. Financ. Econ.* 26 (2), 1871–1885. <https://doi.org/10.1002/ijfe.1883>.
- Leal Filho, W., Ribeiro, P.C.C., Mazutti, J., Lange Salvia, A., Bonato Marcolin, C., Lima Silva Borsatto, J.M., Viera Trevisan, L., 2024. Using artificial intelligence to implement the UN sustainable development goals at higher education institutions. *Int. J. Sustain. Dev. World Ecol.* 31 (6), 726–745.
- Lee, E., Walker, M., Zeng, C.C., 2017. Do Chinese state subsidies affect voluntary corporate social responsibility disclosure? *J. Account. Public Policy* 36 (3), 179–200.
- Li, Q., Shan, H., Tang, Y., Yao, V., Goldstein, I., 2024. Corporate climate risk: measurements and responses. *Rev. Financ. Stud.* 37 (6), 1778–1830.
- Li, Y., Wen, Y., Tao, D., Guan, K., 2019. Transforming cooling optimization for green data center via deep reinforcement learning. *IEEE Trans. Cybern.* 50 (5), 2002–2013.
- Lin, B., Wu, N., 2023. Climate risk disclosure and stock price crash risk: the case of China. *Int. Rev. Econ. Financ.* 83, 21–34. <https://doi.org/10.1016/j.iref.2022.08.007>.
- Lin, J., Zeng, Y., Wu, S., Luo, X., 2024. How does artificial intelligence affect the environmental performance of organizations? The role of green innovation and green culture. *Inf. Manag.* 61 (2), 103924. <https://doi.org/10.1016/j.im.2024.103924>.
- Liu, J., Liu, L., Qian, Y., Song, S., 2022. The effect of artificial intelligence on carbon intensity: evidence from China's industrial sector. *Socio Econ. Plan. Sci.* 83, 101002. <https://doi.org/10.1016/j.seps.2020.101002>.
- Lu, Q., Deng, Y., Wang, X., Wang, A., 2024. The impact of China's green credit policy on enterprise digital innovation: evidence from heavily-polluting Chinese listed companies. *China Fin. Rev. Int.* 14 (1), 103–121. <https://doi.org/10.1108/CFRI-11-2022-0224>.
- Lundvall, B.-Å., Ripak, C., 2022. China's catching-up in artificial intelligence seen as a co-evolution of corporate and national innovation systems. *Res. Policy* 51 (1), 104395.
- Martinez-Jimenez, R., Hernández-Ortiz, M.J., Cabrera Fernández, A.I., 2020. Gender diversity influence on board effectiveness and business performance. *Corp. Gov. Int. J. Bus. Soc.* 20 (2), 307–323. <https://doi.org/10.1108/CG-07-2019-0206>.
- Muen, H.M., Nordin, S., Badru, B.O., 2023. Gender diversity and corporate financial distress in the Pakistan stock market: the interacting effect of family-controlled companies. *J. Fam. Bus. Manag.* 14 (1), 2–27.
- Nazir, T., Nagra, H.F., Bhatti, M.H., Shahid, R.N., Shaikat, N., 2024. Predictive analytics for environmental impact: deep learning and AI ensemble strategies in forecasting CO2 emissions. Available at: <https://assets-eu.researchsquare.com/files/rs-4081410/v1/f56c9548-7844-4ab4-9f4f-f6c067373c3b.pdf>. (Accessed May 2025).
- Nishant, R., Kennedy, M., Corbett, J., 2020. Artificial intelligence for sustainability: challenges, opportunities, and a research agenda. *Int. J. Inf. Manag.* 53, 102104.
- Nordhaus, W.D., 1977. Economic growth and climate: the carbon dioxide problem. *Am. Econ. Rev.* 67 (1), 341–346.
- Nordhaus, W.D., Popp, D., 1997. What is the value of scientific knowledge? An application to global warming using the PRICE model. *Energy J.* 18 (1), 1–45.
- Nuseir, M.T., Al Kurdi, B.H., Alshurideh, M.T., Alzoubi, H.M., 2021. Gender discrimination at workplace: do artificial intelligence (AI) and machine learning (ML) have opinions about it, pp. 301–316. https://doi.org/10.1007/978-3-030-76346-6_28.
- Painter, M., 2020. An inconvenient cost: the effects of climate change on municipal bonds. *J. Financ. Econ.* 135 (2), 468–482.
- Pan, X., Chen, X., Sinha, P., Dong, N., 2020. Are firms with state ownership greener? An institutional complexity view. *Bus. Strateg. Environ.* 29 (1), 197–211. <https://doi.org/10.1002/bse.2358>.
- Petrella, S., Miller, C., Cooper, B., 2021. Russia's artificial intelligence strategy: the role of state-owned firms. *Orbis* 65 (1), 75–100. <https://doi.org/10.1016/j.orbis.2020.11.004>.
- Pham, H.T., Nong, D., Simshauser, P., Nguyen, G.H., Duong, K.T., 2024. Artificial intelligence (AI) development in the Vietnam's energy and economic systems: a critical review. *J. Clean. Prod.* 438, 140692. <https://doi.org/10.1016/j.jclepro.2024.140692>.
- Rana, N.P., Chatterjee, S., Dwivedi, Y.K., Akter, S., 2022. Understanding dark side of artificial intelligence (AI) integrated business analytics: assessing firm's operational inefficiency and competitiveness. *Eur. J. Inf. Syst.* 31 (3), 364–387. <https://doi.org/10.1080/0960085X.2021.1955628>.
- Ren, X., Zhang, X., Yan, C., Gozgor, G., 2022. Climate policy uncertainty and firm-level total factor productivity: evidence from China. *Energy Econ.* 113, 106209.
- Roberts, H., Cows, J., Morley, J., Taddeo, M., Wang, V., Floridi, L., 2021. The Chinese approach to artificial intelligence: an analysis of policy, ethics, and regulation. Ethics, governance, and policies in artificial intelligence 47–79.
- Sautner, Z., Van Lent, L., Vilkov, G., Zhang, R., 2023. Firm-level climate change exposure. *J. Financ.* 78 (3), 1449–1498.
- Singh, A., Kanaujia, A., Singh, V.K., Vinuesa, R., 2024. Artificial intelligence for <sc>sustainable development goals</sc>: bibliometric patterns and concept evolution trajectories. *Sustain. Dev.* 32 (1), 724–754. <https://doi.org/10.1002/sd.2706>.
- Singh, S., Goyal, M.K., 2023. Enhancing climate resilience in businesses: the role of artificial intelligence. *J. Clean. Prod.* 418, 138228. <https://doi.org/10.1016/j.jclepro.2023.138228>.
- Song, Y., Xian, R., 2025. The effect of firm-level climate change risk on tax avoidance: evidence from China. *Appl. Econ. Lett.* 32 (9), 1221–1225.
- Spagnuolo, F., Casciello, R., Martino, I., Meucci, F., 2025. Exploring the impact of artificial intelligence on the pursuit of SDGs: evidence from European state-owned enterprises. *Corp. Soc. Responsib. Environ. Manag.* 32 (2), 1987–2001.
- Su, H., Qu, X., Tian, S., Ma, Q., Li, L., Chen, Y., 2022. Artificial intelligence empowerment: the impact of research and development investment on green radical innovation in high-tech enterprises. *Syst. Res. Behav. Sci.* 39 (3), 489–502. <https://doi.org/10.1002/sres.2853>.
- Todaro, N.M., Testa, F., Daddi, T., Iraldo, F., 2021. The influence of managers' awareness of climate change, perceived climate risk exposure and risk tolerance on the adoption of corporate responses to climate change. *Bus. Strateg. Environ.* 30 (2), 1232–1248. <https://doi.org/10.1002/bse.2681>.

- Toniolo, K., Masiero, E., Massaro, M., Bagnoli, C., 2020. Sustainable business models and artificial intelligence: opportunities and challenges. In (pp. 103-117). https://doi.org/10.1007/978-3-030-40390-4_8.
- Tornatzky, L.G., Fleischer, M., Chakrabarti, A.K., 1990. The Processes of Technological Innovation.
- Tran, L.T.H., Trinh, V.Q., Le, V.T.P., Tu, T.T.K., 2025. Carbon emissions, firm-level climate change exposure, and corporate cash reserves. *Bus. Strateg. Environ.* 34 (5), 5158–5180. <https://doi.org/10.1002/bse.4188>.
- Tseng, C.-J., Lin, S.-Y., 2024. Role of artificial intelligence in carbon cost reduction of firms. *J. Clean. Prod.* 447, 141413. <https://doi.org/10.1016/j.jclepro.2024.141413>.
- Wamba-Taguimdje, S.L., Fosso Wamba, S., Kala Kamdjoug, J.R., Tchatchouang Wanko, C.E., 2020. Influence of artificial intelligence (AI) on firm performance: the business value of AI-based transformation projects. *Bus. Process. Manag. J.* 26 (7), 1893–1924.
- Wan, Q., Tang, S., Jiang, Z., 2023. Does the development of digital technology contribute to the innovation performance of China's high-tech industry? *Technovation* 124, 102738. <https://doi.org/10.1016/j.technovation.2023.102738>.
- Wang, Z., Fu, H., Ren, X., Gozgor, G., 2024a. Exploring the carbon emission reduction effects of corporate climate risk disclosure: empirical evidence based on Chinese A-share listed enterprises. *Int. Rev. Financ. Anal.* 92, 103072. <https://doi.org/10.1016/j.irfa.2024.103072>.
- Wang, Z., Zhang, T., Ren, X., Shi, Y., 2024b. AI adoption rate and corporate green innovation efficiency: evidence from Chinese energy companies. *Energy Econ.* 132, 107499. <https://doi.org/10.1016/j.eneco.2024.107499>.
- Wu, J., Richard, O.C., Triana, M.d.C., Zhang, X., 2022. The performance impact of gender diversity in the top management team and board of directors: a multiteam systems approach. *Hum. Resour. Manag.* 61 (2), 157–180. <https://doi.org/10.1002/hrm.22086>.
- Xiang, X., Li, Q., Khan, S., Khalaf, O.I., 2021. Urban water resource management for sustainable environment planning using artificial intelligence techniques. *Environ. Impact Assess. Rev.* 86, 106515. <https://doi.org/10.1016/j.eiar.2020.106515>.
- Xie, X., Zou, H., Qi, G., 2018. Knowledge absorptive capacity and innovation performance in high-tech companies: a multi-mediating analysis. *J. Bus. Res.* 88, 289–297.
- Zhang, C., Zhou, B., Tian, X., 2022. Political connections and green innovation: the role of a corporate entrepreneurship strategy in state-owned enterprises. *J. Bus. Res.* 146, 375–384. <https://doi.org/10.1016/j.jbusres.2022.03.084>.
- Zhang, R., Hanaoka, T., 2021. Deployment of electric vehicles in China to meet the carbon neutral target by 2060: provincial disparities in energy systems, CO₂ emissions, and cost effectiveness. *Resour. Conserv. Recycl.* 170, 105622.
- Zhao, J., Gómez Fariñas, B., 2023. Artificial intelligence and sustainable decisions. *Eur. Bus. Organ. Law Rev.* 24 (1), 1–39. <https://doi.org/10.1007/s40804-022-00262-2>.